More with Less: The Impact of Mandatory Overtime on Police Wellness and Productivity

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Abstract

Can mandatory overtime provide an effective buffer against labor shortages, or does the resulting work exhaustion prove counterproductive? We study this question by examining a unique policy adopted by a large urban policing agency that mandated overtime by systematically canceling officers' scheduled days off. We leverage this policy's plausibly exogenous variation in a stacked difference-in-differences design to evaluate the causal effect of a single mandatory overtime shift on officer wellness and productivity. Using daily officer-level data, we find that working an additional shift does not significantly affect officer wellness, productivity, or enforcement outcomes. Our findings are precise enough to rule out even modest changes in most outcomes and we find no evidence of differential policy effects across officer demographics or experience. We suggest these null effects are likely due to the modest magnitude of the intervention—a single overtime day followed by rest. Our results suggest that the limited use of mandatory overtime to manage labor shortages carries a low risk for workers in terms of health or productivity, most notably for frontline workers in high-stakes settings like policing.

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1 Introduction

Across the United States, public sector institutions—including schools, hospitals, and police departments—are grappling with severe staffing shortages and challenging work environments (Doan et al., 2024, Maryland Office of Legislative Audits, 2024, Pandemic Response Accountability Committee, 2023). These challenges have been exacerbated by the COVID-19 pandemic and the retirement of an aging workforce. This context is particularly salient for law enforcement agencies, where high vacancy rates have led to increased workloads and concerns about declining officer morale (City of Chicago Office of Inspector General, 2022, Police Executive Research Forum, 2023), further complicated by shifts in public sentiment and tight budgets (International Association of Chiefs of Police, 2020, National Conference of State Legislatures, 2021). The widespread perception, supported by a growing body of literature, is that excessive workloads negatively affect workers. However, the impact of modest and distributed increases in workload—common strategies for managing staffing shortages in law enforcement—is less understood. Determining how these prevalent policies affect the productivity and well-being of public-sector frontline workers is a critically important and unanswered question, especially in high-stakes settings like policing.

Our paper fills this gap by directly examining how short-term, mandatory overtime, particularly that resulting from the cancellation of scheduled days off, affects officer behavior, well-being, and on-the-job performance. Our study focuses on the Chicago Police Department (CPD), which in 2022 introduced a new system of cyclical, pre-scheduled cancellations of Regular Days Off (RDOs) to address significant staffing shortages and meet mandatory staffing requirements. This policy involved canceling the first of an officer's two scheduled RDOs once every six weeks, resulting in a single extended workday followed by a rest day. Crucially, this cyclical system was distinct from traditional event-driven cancellations because it was determined in advance, not tied to specific events, and each cancellation affected hundreds of officers simultaneously. The broad policy and the nuances of its rotation schedule generate plausibly exogenous variation in officer workloads. This variation, unrelated to individual officer behavior or short-term operational needs, allows for an identification strategy to isolate causal effects.

To estimate the effects of these cancellations, we use linked personnel, dispatch, and scheduling data from 2021 to 2023. We identify RDO cancellation events using department-wide schedules and construct a balanced panel of officers with rotating work-rest patterns. Our empirical strategy considers officers with otherwise identical work schedules, comparing officers who have one RDO canceled to others assigned to their usual work schedule. We compare outcomes for officers in the first group to those from the second in the days before and after an RDO cancellation. In particular, we implement a so-called "stacked" event study design with officer, date, and assignment fixed-effects. Our empirical design mitigates concerns about bias related to staggered rollout by excluding early-late comparisons that arise in a two-way fixed-effects specification (de Chaisemartin and D'Haultfoeuille, 2023, Deshpande and Li, 2019, Goodman-Bacon, 2021, Wing et al., 2023). This approach allows us to isolate the causal effect of short-term, externally imposed increases in workload on officer behavior, performance, and wellbeing. This setting offers several advantages: First, it isolates plausibly exogenous, short-run variation in workload, circumventing common confounders related to officer selection for overtime or high-activity shifts. Second, it enables us to track outcomes with high temporal precision. Third, it offers insight into a common policy used by public safety organizations and other frontline agencies to manage chronic staffing shortages.

Leveraging our causal identification strategy, we demonstrate that assignment to an RDO cancellation significantly increases the likelihood of officers working on that RDO. While not a perfect determinant, this outcome confirms broad compliance with the policy and motivates our focus on intent-to-treat effects, as actual compliance may be endogenous. Using this strategy, we find that RDO cancellations do not have a meaningful impact on officer behavior or productivity across a number of key outcomes and do not negatively impact several proxies for wellbeing. In the days following an RDO cancellation, we observe that officers respond to virtually the same number of calls for service and arrive just as quickly; we characterize both of these estimates as "precise" zeros. We also find no change in discretionary enforcement activity, including stops, arrests, and use of force, though these estimates are estimated with less precision. Heterogeneity tests reveal that there are no significant differences in treatment effects between officers by demographic characteristics, enforcement patterns, or the number of RDO cancellations that they previously experienced. Results are robust to a range of modeling choices and we find no evidence of pre-trends or effects on placebo dates. Taken together, the evidence suggests that short-term, externally imposed increases in workload have minimal impact on how officers behave or feel once they are back on the job.

This study contributes to several strands of literature, including those on overtime, burnout, police performance, and public safety management. Much of the existing literature on high-burnout occupations documents the long-term consequences of cumulative fatigue and extreme stressors on worker productivity and turnover (Blackburn et al., 2023, Cho et al., 2023). Research in healthcare and other frontline occupations consistently shows that long hours and stressful conditions can reduce productivity, impair judgment, and increase errors (Blackburn et al., 2023, Lu and Lu, 2017, Nekoei et al., 2024, Sachiko and Isamu, 2016). The policies examined in these studies often differ from the more modest, temporary scheduling adjustments that police leaders commonly resort to. While burnout typically reflects chronic strain, some studies also suggest that short-term increases in workload may contribute to or mimic early burnout symptoms. For example, some descriptive work from the NYPD suggests that overtime correlates with more use of force, complaints, and injuries (New York City Department of Investigation, 2023). This body of evidence suggests that organizational reliance on mandatory overtime as a substitute for adequate staffing may ultimately yield diminishing returns.

The existing literature on overwork is largely descriptive, but a growing number of studies use causal designs to examine its effects. For instance, Linos, Ruffini, and Wilcoxen (Linos et al.) show that peer support can reduce dispatcher burnout, while Fan et al. (2025) found that a four-day work week significantly lowered it. Our project is most closely related to concurrent work by Ferrazares (2025), who also uses data from the Chicago Police Department and leverages a well-accepted identification strategy that exploits idiosyncratic variation in scheduling assignments (Ba et al., 2021, Gudgeon et al., 2024, Holz et al., 2023). Rather than examining a specific policy, Ferrazares examines the cumulative effects of consecutive workdays, finding that each additional day worked increases the risk of use of force and officer injury—a pattern consistent with fatigue. Similarly, experimental evidence from James (2018) finds that fatigued night-shift officers perform worse in simulations. Our study, in contrast, isolates the short-run effect of a policy that imposes a single mandatory overtime day followed by rest, which we might expect to have a different impact than the long-term, cumulative effects of burnout.

Our findings are more consistent with a different strand of management literature that identifies situations in high-stress environments where marginal increases in workloads can be organizationally efficient up to a certain threshold. For example, Ludwig Kuntz (2015) quantify a "tipping point" in hospital mortality, demonstrating that hospitals can absorb marginal patients without substantive harm up to a high occupancy rate. Similarly, Kc and Terwiesch (2009) find that hospitals can increase their service rate under increasing workloads, within limits. Our study contributes to this literature by analyzing a different scheduling margin: the effect of a single mandatory overtime day followed by rest. By exploiting this distinct variation, our study offers complementary evidence that does not contradict findings from the existing literature of an association between excessive work and fatigue. Instead, our precise null effects suggest that fatigue-related effects may depend on sustained exposure without adequate recovery.

Our findings have practical implications for police leaders navigating chronic staffing shortages. While public debate often focuses on sweeping reforms or major recruitment drives, departments typically manage understaffing through marginal adjustments, such as overtime and extended shifts. The Chicago Police Department's systematic and cyclical day-off cancellations exemplify such incremental responses, implemented without the confounding influence of emergent operational needs. This setting reflects the broader reality of high overtime in frontline agencies. For instance, in 2021, Chicago officers already averaged 11 overtime hours weekly on top of a 54-hour schedule. Against this backdrop, our results suggest that modest increases in mandatory overtime do not generate measurable short-term harms—and that equally modest reductions may not yield substantial improvements. While larger structural changes could still affect burnout or morale, our evidence indicates that the marginal adjustments most departments rely on are unlikely to meaningfully shift outcomes. These findings may also extend to other public-sector workforces, where overtime is a routine stopgap, particularly frontline workers.

2 Institutional Context, Data, and Sample Construction

2.1 Institutional Context

The Chicago Police Department (CPD) provides a useful setting to study the impacts of mandatory overtime. Like many jurisdictions in the United States, CPD experienced gradual declines in the number of sworn officers over the last two decades. These declines became especially acute between 2019 and 2022, when retirements and resignations among sworn Chicago police officers nearly doubled, producing a 10% reduction in patrol officers. Although it is difficult to pinpoint the specific causes of these departures, they coincided with the upheaval of the COVID-19 pandemic and a wave of nationwide protests against excessive use of force by police.

In response to these staffing shortages and the protests that followed the killing of George Floyd, CPD began canceling officers' scheduled days off in May 2020. The goal was to ensure sufficient staffing during periods of heightened crime and disorder. Over the subsequent three years, CPD continued to rely on day-off cancellations, effectively mandating overtime across the force. In February 2022, the department formalized this practice by instituting pre-determined cancellations that cycled through groups of officers. Importantly, officers had little choice over when these mandatory days off occurred.[1] Concerns about the toll on staff soon surfaced. A 2022 audit revealed that some officers worked as many as 12 consecutive days with shifts as long as 12 hours in April and May of that year (City of Chicago Office of Inspector General, 2022). The policy remained in place until October 2023, when newly appointed Superintendent Larry Snelling announced a general prohibition on mandatory overtime, limiting it only to holidays and special events (Sisk, 2023).

The increasing reliance on overtime is visible in CPD's administrative data. As shown in Appendix Figure C.7, the average number of days officers worked overtime rose by 29% between 2019 and 2023. Relative to 2014—the earliest year for which data are available—overtime days were 44% higher in 2023. While voluntary overtime has steadily grown in recent years, there is also a clear increase in involuntary overtime beginning in 2020, consistent with CPD's evolving policy. For 2022 and 2023 specifically, regular day-off (RDO) cancellations and mandatory shift extensions contributed roughly 17% and 9% of total overtime days, respectively. These figures underscore both the scale and institutionalization of mandatory overtime during our study period.

Understanding CPD's work assignment system is critical for interpreting the overtime policy. Each officer is assigned to a "day-off group" (DOG) that follows a repeating six-day cycle: four consecutive work days followed by two consecutive rest days. DOG assignments are determined through a seniority-based bidding process in the preceding fall. Although officers can and do change DOGs throughout the year—we observe switching in nearly 40% of officer-years—they ultimately lack direct control over their schedules, which remain subject to managerial discretion. In addition to DOGs, CPD assigns officers to a geographic and unit placement. The city is divided into five patrol areas, which are further split into 22 districts and then into 277 beats. Officers may also be assigned to non-patrol units that operate outside of district boundaries. New recruits are initially assigned to a district and unit, while experienced officers can bid for changes based on seniority.

Mandatory overtime was operationalized through "cancellation matrix" calendars that determined when RDOs would be revoked. Each cycle began by identifying a subset of areas, and within those areas, cancellations cycled sequentially across DOGs. Subsequent cycles rotated to different areas and repeated the process. This rotating system meant that, while officers working in different districts and units varied in many respects, the timing of their mandatory overtime

[1]Additional mandatory overtime was also assigned during special events and holidays independently of this policy.

was driven by a departmental schedule rather than officer choice. As a result, the policy induced plausibly exogenous variation in which officers were required to work on any given day off.[2]

The structure of DOGs and the rotating cancellation matrix underpin our identification strategy. Specifically, we exploit the cyclical day-off cancellation pattern implemented in 2022 to construct a stacked difference-in-differences design. Because officers' exposure to mandatory overtime is determined by their DOG assignment and the department's rotating cancellation schedule—rather than officer choice—it provides plausibly exogenous variation at the officer level. This institutional design allows us to examine the causal effects of mandatory overtime on officer outcomes.

2.2 Data

Our analysis uses administrative data obtained through Freedom of Information Act (FOIA) requests and public sources. Officer attendance data come from the CPD's Attendance and Assignment (A&A) database, which records daily presence, absence reasons, and shift/unit/beat assignments. We separately observe day-off group (DOG) assignments, and obtained DOG work schedules from operations calendars.

We measure treatment by combining these schedules with RDO cancellation dates from CPD cancellation matrices. We identify an officer as being treated when the RDO for their DOG-Area combination is canceled. We verify compliance using A&A records and officer-day overtime data, which distinguish voluntary from mandatory hours. We measure compliance with a binary indicator for reporting to work on a canceled RDO and a continuous measure of overtime hours worked.

We study outcomes in three domains: officer wellness, productivity, and enforcement. Each captures a distinct channel through which mandatory overtime may affect officers and the public. We measure wellness using absence reasons reported in the A&A data. Specifically, we construct daily indicators for whether an officer (1) was absent due to an injury sustained on the job, and (2) called in sick. We also consider attendance and overtime in the week following a canceled RDO as complementary measures of wellness. Together, these outcomes capture whether mandatory overtime increases short-term illness or injury risk, or whether officers make their own workload adjustments to mitigate the immediate effects of the policy.

[2] Recent research has also used CPD's DOG system to study how officer race, gender, and experience shape performance (Ba et al., 2021, Gudgeon et al., 2024).

We measure productivity by counting the number of incidents involving each officer, either through responding to a call for service (CFS) or by initiating an investigatory stop. CFS data capture incidents for which a 911 dispatcher requested and received a police response, and thus reflect responses to public demand for police assistance. Investigatory stop data capture officer-initiated encounters in which an officer temporarily detains an individual suspected of criminal activity. We obtained universe files for both incident types from CPD and construct daily officer-level counts of CFS responses and investigatory stops.[3] Because both activities involve some degree of discretion (less in CFS, more in investigatory stops), mandatory overtime may reduce engagement if overburdened officers choose to shirk.

Our third domain—enforcement activity—is closely related to productivity but conceptually distinct. Whereas productivity measures the extent of public engagement, enforcement measures the intensity of those interactions. Enforcement decisions are often made in high-stakes, subjective contexts where overworked officers may be more likely to escalate. We use two daily officer-level indicators to capture these outcomes: whether the officer was involved in an arrest and whether the officer used force. Arrest data provide the universe of CPD arrests with information on charges and all officers involved (primary, secondary, or assisting). Because arrests are relatively rare at the officer-day level, we define an arrest indicator equal to one if an officer contributed to any arrest that day. Use-of-force data identify officers who reported using physical force. CPD requires officers to document force used against "actively resisting" civilians, resulting in reported injury, or involving civilians who threatened, obstructed, or assaulted an officer (Chicago Police Department, 2023).[4]

We supplement these data with CPD roster files containing officer demographics and tenure. To link officers across datasets, we construct a unique identifier using names and appointment dates, matched to badge numbers, DOG assignments, and A&A records. This produces an officer-day panel covering 93% of the 13,601 officers in the roster. Linkage rates are high: 96.5% of overtime data, 94% of use-of-force incidents, 90% of investigatory stops, 89.7% of arrests, and 91% of CFS. The final dataset is a unified officer-day panel combining treatment exposure, demographics, and outcomes across the three domains. This structure provides

[3] CFS volume is top-coded at the 99th percentile (21 calls in a single day). We also measure the time between dispatcher receipt of the CFS and officer arrival on scene, included as a supplementary outcome in Figure C.9a.

[4] In 2022–23, 79% of reports indicated use of control tactics (e.g., holds, restraints); 61.7% reported bodily force without weapons (e.g., punches, tackles); and 4.4% reported use of a weapon (e.g., tasers, batons, firearms).

broad coverage and allows us to examine both average effects and heterogeneity in officer responses.

2.3 Sample Construction

Building on the officer-day data described above, we restrict the sample to officers and schedules directly shaped by the CPD's mandatory overtime policy We analyze officer-date observations from May 31, 2022 through October 20, 2023, covering the period when the RDO cancellation policy was fully in effect.[5] Because cancellations were administered through the DOG system, we focus on patrol officers in six DOGs (61–66) whose schedules followed a stable four-days-on, two-days-off rotation. These groups were sequentially offset by one day, ensuring consistent coverage across the week. Restricting to these DOGs provides comparability across officers and aligns treatment precisely with institutional practice. We further restrict to units assigned to the five main patrol areas (Areas 1–5). Citywide units, which operate across multiple areas, were excluded because they did not follow the same cancellation logic. To minimize noise from short-term assignment changes, we impute DOG and area by officers' modal assignment within each month. Cancellations always fell on the first rest day of the cycle. Complying officers therefore worked on that day, rested the next, and then restarted their cycle. We use this structure to construct a series of "sub-experiments": for each cancellation, we compare officers in the same DOG but assigned to different areas, only some of whom experienced the cancellation. This within-DOG, within-cycle design ensures that treatment exposure is driven by the rotation of the cancellation matrix, rather than officer choice.

Our main sample follows officers for five days before and after a cancellation, capturing the work cycles directly surrounding treatment. This window allows us to measure immediate changes in wellness (injuries, sick leave, attendance), productivity (calls for service and investigatory stops), and enforcement (arrests and use of force). For each sub-experiment, we construct balanced panels, excluding officers we do not observe in all periods. We also exclude officers who were treated within the ten-day window around the focal cancellation.[6]

Pooling across 215 observed cancellations yields a stacked sample covering roughly two-fifths of patrol officer-dates during the study period.[7] In this sample,

[5]We also observe cancellation matrices for February–March 2022, but assignment patterns during those months were inconsistent with the formal policy. By May 31, cancellation patterns match the expected schedules.

[6]We provide results from longer- and alternative short-run windows in the Appendix.

[7]The full linked data cover 7,566 officers and 2.4 million officer-days. The stacked sample consists of 5,606 officers and 1.2 million officer-days. Demographics and baseline outcomes are

38% of officer-dates fall into the treatment group, though only about 2% of all officer-dates include involuntary overtime hours, reflecting incomplete compliance. As shown in Table 1, absences are somewhat common (on an average day, 4.5% of officers call out injured, 6.5% sick, and 9.4% take paid time off), and officers change assignments infrequently (2% DOG, 1% area). Officers average 2.4 responses to calls for service per day, initiate 0.06 investigatory stops, make 0.03 arrests, and use force on 0.16% of days. These outcomes provide the empirical foundation for assessing how mandatory overtime reshaped wellness, productivity, and enforcement.

3 Empirical Design

Our empirical design relies on the quasi-random assignment of officers to overtime introduced by RDO cancellations. The cyclical pattern of these cancellations, combined with DOG and area assignments for each officer, generates within-DOG and between-area variation. In effect, our design compares officers in the same DOG who would otherwise have identical schedules, but differ due to RDO cancellations in some areas. As described in section 2.2, we consider each RDO cancellation as a single natural experiment induced by the overtime policy. By pooling data across sub-experiments in a stacked sample, a stacked difference-in-differences design allows us to leverage all RDO cancellations we observe to estimate the effects of mandatory overtime on a range of officer-level outcomes. This design avoids problematic comparisons between early and later treated observations that can arise in a two-way fixed-effects specification in the presence of treatment heterogeneity (de Chaisemartin and D'Haultfoeuille, 2023).

To explore potential temporal dynamics in our outcomes and to assess pre-treatment trends across groups, we estimate event study models of the following form:

$$\begin{aligned} Y_{iet} = {} & \alpha_1 D_{ie} + \sum_t (\beta_t T_t + \delta_t D_{ie} \times T_t) \\ & + \alpha_2 DaysLastTreated_{iet} + \phi_i + \eta_e + \kappa_{ie}^{A \times d} + \kappa_{ie}^{D \times d} + \epsilon_{iet} \end{aligned} \tag{1}$$

Here Y_{iet} reflects the outcome of interest for officer i in sub-experiment e at event time t. D_{ie} is officer i's treatment assignment in sub-experiment e – recall that this treatment assignment will vary within officers across sub-experiments

nearly identical between the full and stacked samples, aside from a slightly lower share of Black officers (−1.4%). Detailed balance tables are in Table 1.

as officers not called to work on a given RDO can be in the treated group in a future or past sub-experiment. We include a set of event time indicators, T_t, and treatment by event time interactions $D_{ie} \times T_t$. We exclude the event time indicator and interaction for the day before a cancellation ($t = -1$), so the parameters are estimated relative to the difference in treatment and control at event time $t = -1$. We also control for $DaysLastTreated_{iet}$, the number of days since an officer was last treated and include officer (ϕ_i), area - by - date ($\kappa_{ie}^{A\times d}$), and DOG -by- date ($\kappa_{ie}^{D\times d}$) fixed-effects.

The set of δ_t on the interactions of treatment and event time are our coefficients of interest. We consider officers who change DOG-area assignments within a given month to be non-compliers, and therefore interpret the δ_t as intention-to-treat (ITT) estimates. To estimate an aggregated δ across the post-periods of interest, we modify equation 1 to include only a single post-treatment indicator and interaction and omit any observations that fall on an RDO, including data from relative times -5, 0, and 1. We do this to better capture the effect of assignment to RDO cancellation on workday outcomes: we do not expect to observe changes in work performance on an officer's day off, and including these dates complicates interpretation of our results.

Following Wing et al. (2023), we specify how our estimator aggregates treatment effects across sub-experiments. We apply observation-level weights so that our estimates, $\hat{\delta}_t$, correspond to weighted averages of sub-experiment ITTs, with weights equal to each sub-experiment's sample share, shown in 2.

$$\delta_t = \sum_{\Omega_e} ITT(e,t)\frac{N_e}{N} \tag{2}$$

N_e is the number of total observations in sub-experiment e, N is the size of the total stacked sample, and $\boldsymbol{\Omega}_e$ is the universe of sub-experiments. To achieve this estimand, we weight treatment and control observations as follows in 3, where N^g denotes the number of observations in the group (treatment or control) to which the weights will apply:

$$\frac{N_e/N}{N_e^g/N^g} \tag{3}$$

Because treatment assignment is determined by the combination of an officer's area and DOG assignments, we cluster standard errors at the area-DOG level.[8]

Our empirical strategy enables us to avoid key sources of endogeneity in the relationship between overtime and worker performance. First, because overtime is normally voluntary, individuals who work extra hours might systematically differ from those who do not. We overcome this selection bias by examining mandatory overtime imposed through RDO cancellations; our intention-to-treat (ITT) estimates are robust to imperfect compliance. Second, in law enforcement, the demand for overtime often correlates with spikes in crime, potentially confounding results. We mitigate this by leveraging the CPD's predetermined, sequential RDO cancellation pattern, which is largely orthogonal to immediate crime trends or demand fluctuations. Third, we use a daily officer panel within a difference-in-differences framework, incorporating officer, date, and assignment fixed-effects, to account for non-random officer selection into specific assignments, areas, and initial level differences in working conditions. This comprehensive approach allows us to isolate the causal effect of short-term, externally imposed increases in workload on officer behavior, performance, and well-being.

3.1 Identifying Assumptions

Our identification strategy relies on several standard assumptions. The "no anticipation effects" assumption requires that officers do not change behavior in advance of a cancellation. Institutionally, the CPD's rigid seniority-based annual scheduling limits opportunities for officers to make anticipatory changes to their work schedules before cancelled RDOs. Empirically, our event study plot in Figure 1b confirms that there is no evidence of anticipation effects for absences. In Appendix section C.2, we also show that officers do not differentially change DOG or area to evade treatment. Next, the parallel trends assumption requires that treatment and control group outcomes would evolve similarly without the canceled RDO. This is plausible given our observation of comparable pre-treatment trends for all outcomes; we present event study estimates for the four regular work days preceding treatment for each outcome in section 4. While we are powered to detect minor variations in pre-treatment event study estimates, these statistical violations are best characterized as idiosyncratic or small and do not amount to meaningful evidence of differential pre-trends that would invalidate our inference.

Furthermore, because CPD implemented cancellations sequentially and across the entire city, it is unlikely that our results are driven by localized shocks or

[8]Alternative standard error assumptions yield qualitatively similar results (available upon request).

unexpected events that happened to overlap with treatment. An alternative concern is that the broad scope of officers affected by cancellations allows our control groups to include officers who were treated in other sub-experiments. However, this does not introduce bias if treatment effects are short-lived – specifically, that they do not persist into subsequent work cycles after our evaluation window. This is reasonable in our institutional setting, where mandatory overtime occurs no more than once a month and each work cycle with a canceled RDO is followed by two days off. In addition, our regressions explicitly control for the number of days since an officer was last treated. As a robustness check, we relax this assumption and allow treatment effects to last for two full work cycles instead of one. The results are highly similar under this alternative, as shown in section C.4 and section C.5.

4 Results

4.1 Treatment Compliance

In this paper, we estimate the effects of overtime on officer well-being, productivity, and enforcement outcomes using RDO cancellations as conditionally quasi-random variation in days worked. Because officers may switch assignments or take alternative leaves throughout the year, we neither expect nor observe perfect attendance – or perfect non-attendance – among the treatment or control groups. [9] However, we can clearly demonstrate the validity of the first stage for producing intent-to-treat estimates: on average, officers do work more when their RDO is canceled.

We begin by showing evidence of this pattern in the raw data. Based on CPD operations calendars, we expect officers to work four consecutive days and then rest for two before restarting the work cycle. In event time, these RDOs fall on days -5, 0, and 1. In Figure 1a, we plot the proportion of treatment and control officers reporting for work in each day of our sub-experiments. Treatment and control officers are equally likely to report for work on scheduled work days ($t = -4, ..., -1$ and $t = 2, ..., 6$), and are roughly 50 percentage points less likely to report for work on a scheduled and granted day off ($t = 1$). However, while control group attendance rates drop similarly on the canceled RDO ($t = 0$), about half of treatment group officers report to work. This corresponds to treatment officers attending work at a rate approximately 25 percentage points higher than control officers on the date of treatment.

[9] Importantly, as we describe in section 3 and show in section C.2, assignment switching is not an anticipatory response to a canceled day off.

This estimate is virtually unchanged when we formalize the comparison in our event study specification. In Figure 1b, we estimate that being assigned treatment statistically significantly increases an officer's probability of coming into work on the day of the cancellation by over 25 percentage points. That the ratio of officers calling out sick to those present at work is stable between the pre-treatment periods and the canceled day off (visible in Figure 2c) further affirms a high degree of compliance on the extensive margin.

On the intensive margin, Figure 1c and Figure 1d display binscatter and event-study plots, respectively, of the number of overtime hours an officer works on a given day. Both plots show that, on canceled RDOs, treatment officers work an average of approximately three additional hours compared to the control group. This is roughly equal to the length of a standard 9-hour shift multiplied by the compliance rate, implying that complying officers work a full shift on the canceled RDO. In total, we find strong evidence that a canceled RDO induces a real, short-term increase in officer work time.

4.2 Attendance and Officer Wellness

A primary criticism of mandatory overtime, and a key factor behind the CPD's decision to discontinue canceling RDOs, is the concern that it undermines officer wellness. This may be particularly relevant for physically demanding and public-facing jobs such as patrol work. To evaluate this possibility, we use detailed attendance records to examine whether canceled RDOs affect direct measures of wellness—such as illness or injury absences—as well as compensatory behaviors like reducing subsequent time at work. Panel A of Table 2 reports these results.

The raw data provide little evidence of compensatory behavior. In Figure 1a, treatment and control officers display nearly identical attendance rates on regular work days, but diverge sharply on the canceled RDO itself, consistent with treatment assignment. This divergence does not persist into subsequent workdays, indicating that treatment officers are no more likely than controls to take additional days off. Similarly, Figure 1c shows no decline in overtime hours in the days following a canceled RDO. Turning to absences, Figure 2a indicates that injury-related absences are highly stable across groups and time, while sick day rates in Figure 2c fluctuate more but maintain parallel differences between treatment and control officers.

Event study estimates confirm these patterns. As shown in Figure 1b and Figure 1d, treatment and control groups exhibit parallel pre-trends and no evidence of compensatory behavior after the canceled RDO. Likewise, Figure 2b and

Figure 2d show that injury and sick absences remain flat and do not deviate meaningfully from pre-treatment trajectories in the post-period.

The aggregated difference-in-differences estimates in Panel A of Table 2 provide precise evidence of null effects. Officers do not compensate for canceled RDOs by missing subsequent workdays (–0.35 pp, insignificant) or by reducing their overtime hours (a negligible increase of 2.4 minutes per day). Nor do we observe any significant increase in absences due to injury or illness: injury absences rise by only 0.04 pp (about 1 percent of the pre-period mean), while sick absences actually decline by 0.18 pp (insignificant). The confidence intervals for these estimates rule out economically meaningful increases, suggesting that mandatory overtime does not compromise officer attendance or health.

4.3 Officer Productivity

Although officers continue to show up for work following a canceled RDO, mandatory overtime may still affect how productive they are once on duty. To test this possibility, we examine two central measures of patrol productivity: calls for service (CFS), which capture responses to public demand, and investigatory stops, which capture discretionary officer activity.

The binscatter plots in Figure 3a and Figure 3c show that treatment and control officers had nearly identical pre-treatment patterns in both outcomes. The one exception is a slight divergence in investigatory stops just prior to treatment, but this difference is economically trivial, amounting to fewer than 0.01 stops per officer per day.

Dynamic event study estimates, presented in Figure 3b and Figure 3d, confirm that pre-treatment trends are parallel and that treatment has no post-period effects. The daily treatment effects are small and stable across the entire work cycle following the canceled RDO.

Aggregated difference-in-differences estimates, reported in Panel A of Table 2, columns 5 and 6, corroborate these findings. Assignment to a canceled RDO has no statistically or economically significant effect on productivity: officers respond to 0.003 additional calls per day (95% CI: –0.061, 0.068) and initiate 0.002 fewer stops per day (2.3 percent of the mean, with the 95% CI ruling out decreases larger than 0.007). These results provide strong evidence that mandatory overtime does not affect the quantity of officers' on-the-job activity.

4.4 Enforcement Activity

Finally, we examine whether canceled RDOs affect officer enforcement behavior, specifically arrests and the use of force. These outcomes represent a more consequential margin of officer behavior and are directly linked to public safety.

The raw data suggest little evidence of systematic differences between treatment and control officers. As shown in Figure 4a and Figure 4c, about 10 percent of officers make at least one arrest on a given workday, while about 0.2 percent use force. These rates and trends are similar across treatment and control groups prior to a cancellation, with the exception of an idiosyncratic fluctuation in use-of-force at $t = -3$. As noted in 3, this appears to be noise in a single period rather than evidence of a violated parallel trends assumption.

The dynamic event study plots in Figure 4b and Figure 4d show no consistent differences between treatment and control groups in either arrests or use of force. Pre-treatment trends track closely, and post-treatment estimates remain flat, indicating that mandatory overtime does not affect the likelihood of enforcement actions.

Aggregated difference-in-differences estimates, presented in Panel A of Table 2, columns 7 and 8, confirm these results. Officers assigned to a canceled RDO are 0.16 pp more likely to make an arrest in the post-period, an effect that is statistically insignificant and corresponds to just 1.7 percent of the pre-treatment mean. The probability of using force falls by 0.01 pp, also statistically insignificant. Disaggregated results by arrest type, reported in Table 3, show no evidence of substitution across offense categories. Insignificant changes range from 1.6 percent to –3.9 percent of baseline means for index, traffic, and other arrests. The only large proportional effect arises for arrests without charges, which are rare (0.4 percent of officer-days). Here, the point estimate suggests a 0.04 pp increase, or 10.5 percent of the mean, but given the low base rate, we regard this as economically negligible. Overall, the evidence indicates that mandatory overtime does not alter officers' enforcement practices.

4.5 Robustness

This section demonstrates that our main findings are robust to alternative specifications and sampling choices. We begin by varying the fixed effects structure. Because RDO cancellations are imposed at the DOG–area–date level, our baseline specification includes officer, DOG-by-date, and area-by-date fixed effects. As shown in Table B.4, the results are unchanged when simplifying the model by dropping officer fixed effects, and they remain stable under a more conservative specification that additionally includes DOG-by-area fixed effects.

We next consider the possibility that treatment effects persist for longer than a single work cycle. If so, our control group could be contaminated by recently treated officers, attenuating our estimates. To address this concern, we re-estimate our models using a restricted sample that excludes officers treated within the prior two work cycles. The results, reported in Table B.6 and Table B.7, are substantively similar to our baseline specification and generally characterized by small, insignificant effects across domains. The only exceptions are modest increases in investigatory stops and "other" arrests, though even these are small in absolute terms. The corresponding event study plots in Appendix section C.4 confirm that these findings are consistent over time.

Finally, we examine whether treatment effects might emerge with a delay, in which case our short post-treatment window could understate the true policy impact. To test this, we extend the event window to include two full work cycles before and after each cancellation, restricting to officers not treated within that horizon. As reported in Table B.8 and Table B.9, the difference-in-differences estimates for attendance, wellness, productivity, and enforcement remain uniformly small and statistically insignificant. Appendix section C.5 presents the associated binscatter and event study plots, which corroborate the null results and show that temporal aggregation does not mask meaningful dynamics across periods, aside from a modest decline in enforcement in the second work cycle.

Together, these robustness checks confirm that our findings are not sensitive to fixed effects structure, assumptions about the persistence of treatment effects, or the length of the post-treatment window.

4.6 Treatment Heterogeneity

In the preceding sections, we find evidence that RDO cancellation does not affect officer engagement across a range of measures. However, it's possible that officers with different characteristics experience different effects that are obscured when averaging over all observations. This possibility is both theoretically and empirically grounded. A large literature documents how worker productivity varies with experience, broadly, due both to selection and learning-by doing (Haggag et al. (2017), Daveri and Parisi (2015), Papay and Kraft (2015)), and police productivity specifically differs by demographic and tenure characteristics (Ba et al. (2021), Ba et al. (2022)). Under mandatory overtime, workers more susceptible to burnout are less able to opt out of the additional hours. For this reason, it is useful to characterize officer traits that predict larger effects. We test for heterogeneity in the effects of mandatory overtime over a range of officer characteristics including demographics, experience, productivity, and history of canceled RDOs. To do so, we split the sample by officer characteristics and estimate the prior difference-in-differences specification using each of the resulting 11 sub-samples. We present the estimated treatment effects for each subgroup in Figure 5 and Figure 6. For example, we show the effect of being assigned RDO cancelation on officer attendance for the subsample of male, and the subset of female, officers separately.

Officer demographics are directly reported in rosters. Due to small sample sizes, we do not include results for officers who identify as Asian, Native/Pacific Islander, or Other. We define the categories of officer experience, tenure, and treatment history based on the median officer in our sample. We label officers who have worked fewer years than the median officer as "inexperienced"; officers who responded to fewer calls for service than the median as "unproductive"; and officers who are treated fewer than 13 times (approximately the median) as being "treated fewer times".

In Figure 5, we plot the coefficients and 95% confidence intervals from the split samples for each of our attendance and wellness outcomes: attendance, overtime hours, injuries, and sick days. Broadly, results across subgroups and outcomes support the null results from our primary analyses. No group exhibited statistically significant or economically large changes in the probabilities they reported to work, called out injured, or called out sick. Echoing our primary results, we find small, positive point estimates for effects on overtime hours for all groups – most, but not all of which are statistically significant. While we observe some variability in the point estimates across groups, we nonetheless observe considerable parity in point estimates for natural comparison groups: when comparing effects for male to female officers, White to Hispanic and Black officers, etc. we find qualitatively similar results. The rare exception to this pattern is in overtime hours, where officers treated fewer times exhibit a noticeably larger increase in overtime hours than their more-often treated counterparts.

Figure 6 presents subgroup treatment effects for enforcement and productivity-related outcomes. Across these domains, our estimates are somewhat noisier than the attendance and wellness results, but show similar trends: we observe small, statistically insignificant point estimates for nearly all outcomes and subgroups, with results congruent across subgroups. Again, we note one exception to this pattern, this time in the "Any Arrests" outcome. Whereas Hispanic officers show a moderate increase in arrest propensities following treatment, Black officers exhibit a moderate decrease. Because this finding is not consistent with a broader pattern of responses for Hispanic or Black officers, we are inclined to attribute it to happenstance. Officers of various demographic backgrounds, experience levels, productivity patterns, and treatment histories demonstrate stable wellness, workplace engagement, productivity, and enforcement outcomes after mandatory overtime.

5 Conclusion

In this paper, we evaluate the effects of a mandatory overtime policy on patrol officer health and performance, studying a unique policy implemented by the Chicago Police Department (CPD) from 2022 to 2023. This policy addressed chronic staffing shortages by systematically canceling officers' scheduled days off in a six-week rotation, thereby increasing their work hours. Using a stacked difference-in-differences design, which leverages plausibly exogenous variation in these cancellations , we find robust evidence that the policy had no significant effect on a broad range of outcomes.

Our analysis shows that a single mandatory overtime shift does not meaningfully affect officer wellness, productivity, or enforcement activities. We find no evidence that the policy led to negative health effects, such as an increase in injury or illness-related absences, and our estimates are precise enough to rule out even modest changes in most outcomes. Similarly, we find no detrimental effects on the public, as officers' productivity—measured by their response to calls for service and investigatory stops—remained unchanged.

We offer two potential explanations for these precise null effects, which likely operate in tandem. First, the magnitude of the intervention was modest. The policy imposed a single mandatory workday, resulting in a five-days-on, one-day-off cycle for affected officers. At an individual level, officers were assigned to mandatory overtime no more than once a month. While it is reasonable to anticipate that sufficiently large or sustained workload increases could alter officer behavior and well-being, our findings do not preclude the existence of such a "tipping point" for fatigue or burnout. Instead, they suggest that temporary increases in workload, particularly those followed by a rest day, may not be sufficient to induce a measurable decline in performance. We study only the short-run effect of this policy and note that it is possible that there are deleterious effects that build over the long-run.

Second, the characteristics of the work performed on these overtime days may have contributed to our results. While our data show that officers are not any less likely to engage in high-intensity tasks like responding to calls for service or making arrests than on a typical workday, we cannot definitively rule out that their work activities are different in unobservable ways. It may be possible that supervisors preemptively assign or that officers self-select into lower-intensity duties on these extended shifts. This form of substitution could be a key mechanism through which the policy avoids detrimental effects on officers and the public.

In summation, our findings suggest that a limited mandatory overtime policy can be a useful short-run tool for police departments and other organizations facing significant labor constraints. Because we find precise null effects, our research also indirectly suggests that equally modest reductions in workload are unlikely to produce substantial improvements in frontline worker performance or well being. For public service provision, decisions regarding marginal workload adjustments may be better evaluated on the basis of aggregate changes and organizational needs rather than the expectation of meaningful changes in individual employee optimization.

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6 Tables

Table 1: Summary Statistics

	Linked Sample		Stacked Sample	
VARIABLES	Mean	SD	Mean	SD
Reports to Work	.5382	.4985	.5455	.4979
Total Overtime Hours	.9277	2.551	.7835	2.305
Total Involuntary Overtime Hours	.1952	1.298	.1636	1.198
Total Calls for Service	1.88	3.512	2.373	3.856
Total Investigatory Stops	.04466	.316	.05971	.3663
Any Arrests	.0587	.2351	.07468	.2629
Any Use of Force	.0012	.0356	.001749	.04287
Female	.2037	.4028	.2202	.4144
White	.6845	.4647	.6838	.465
Hispanic	.346	.4757	.3872	.4871
Black	.1477	.3548	.1649	.3711
Years of Experience	12.29	8.869	9.843	8.053
N	3,152,660		1,854,336	
Injured on the Job	.03508	.184	.04412	.2054
Calls in Sick	.05634	.2306	.06146	.2402
N	3,147,027		1,850,441	
Distinct Officers	7,927	-	5,864	-
Distinct Sub-Experiments	-	-	288	-
Officers per Sub-Experiment	-	-	6,857	1,614

Table 2: Main Outcomes

	(A) Attendance and Work		*(B) Officer Wellness*	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
Treatment × Post	-0.0035	0.0433	0.0004	-0.0018
	(0.0040)	(0.0142)	(0.0009)	(0.0019)
Pre-Period Mean	0.6735	0.5040	0.0444	0.0671
Observations	1348608	1348608	1345779	1345779
	(C) Officer Productivity		***(D) Enforcement Activity***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
Treatment × Post	0.0034	-0.0018	0.0016	-0.0001
	(0.0328)	(0.0027)	(0.0023)	(0.0002)
Pre-Period Mean	3.0738	0.0779	0.0965	0.0021
Observations	1348608	1348608	1348608	1348608

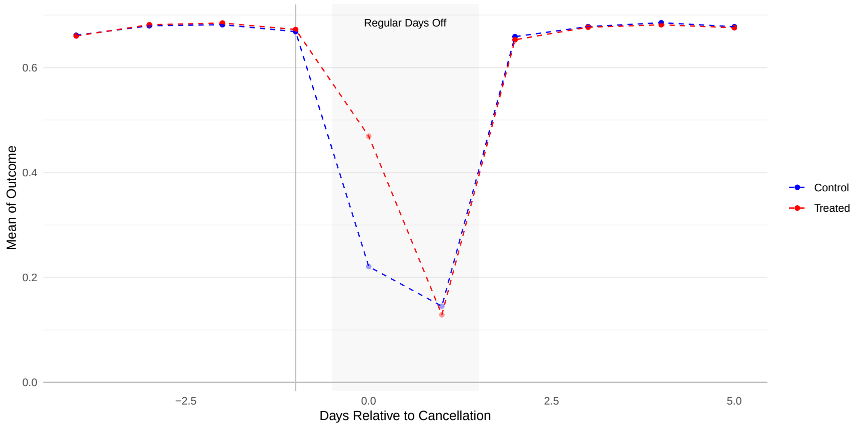
Data are restricted to exclude observations falling on RDOs, specifically with event-times $t \in \{-, 0, 1\}$. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table 3: Arrest Type

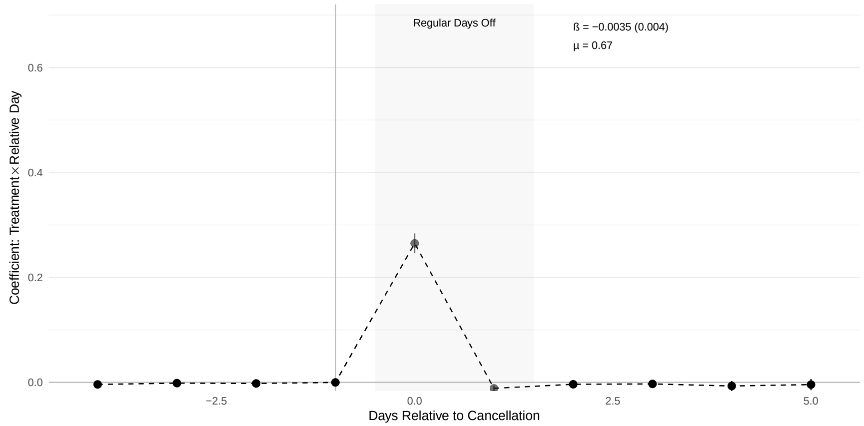
	(1)	(2)	(3)	(4)
Dependent Variable:	Any Index Arrests	Any Traffic Arrests	Other Arrests	Any Arrests with No Charge
Treatment × Post	-0.0006	-0.0011	0.0013	0.0004
	(0.0011)	(0.0015)	(0.0018)	(0.0006)
Pre-Period Mean	0.0196	0.0283	0.0837	0.0038
Observations	1348608	1348608	1348608	1348608

Data are restricted to exclude observations falling on RDOs, specifically with event-times $t \in \{-, 0, 1\}$. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

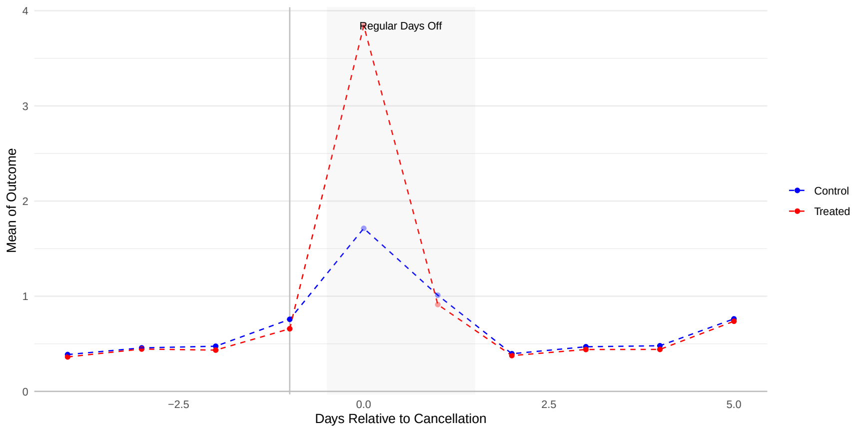
7 Figures



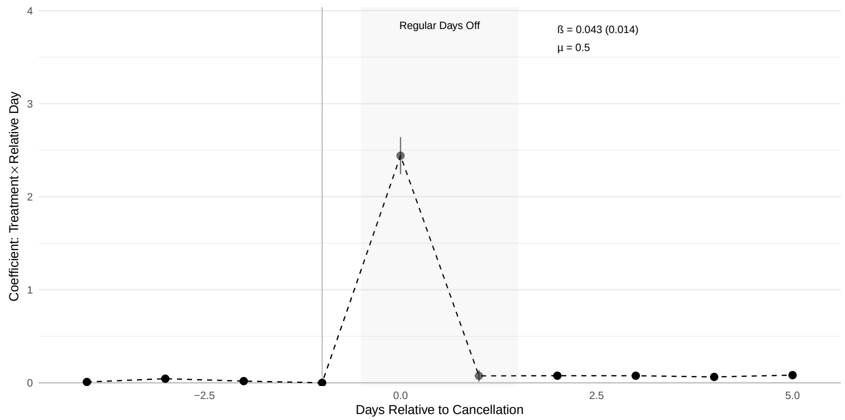
(a) Raw Data: Reported to Work



(b) Event Study: Reported to Work

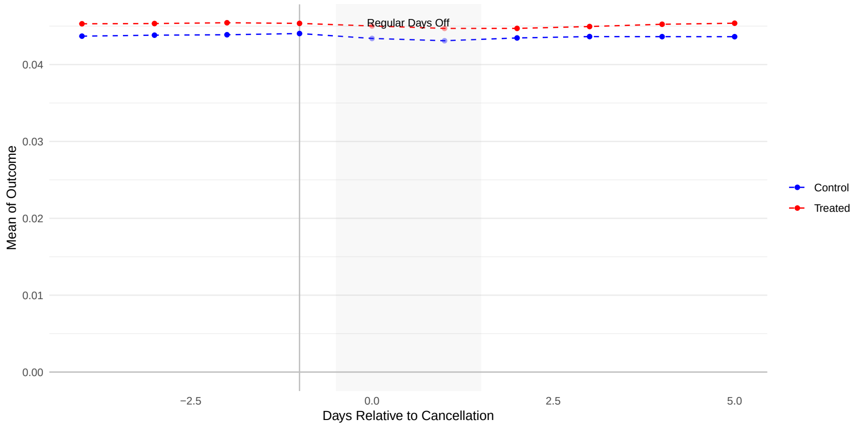


(c) Raw Data: Overtime Hours

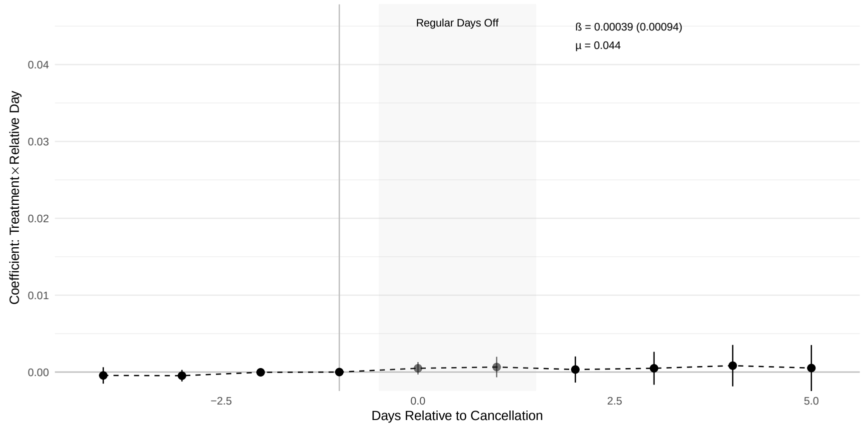


(d) Event Study: Overtime Hours

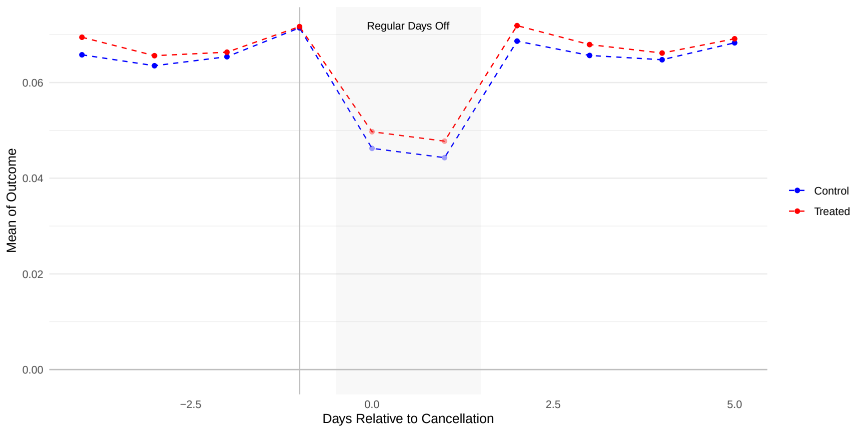
Figure 1: Treatment Compliance and Effects on Attendance



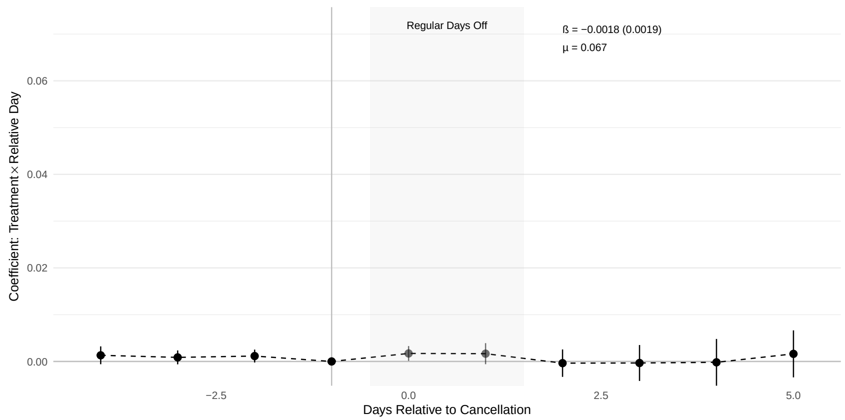
(a) Raw Data: Injury



(b) Event Study: Injury

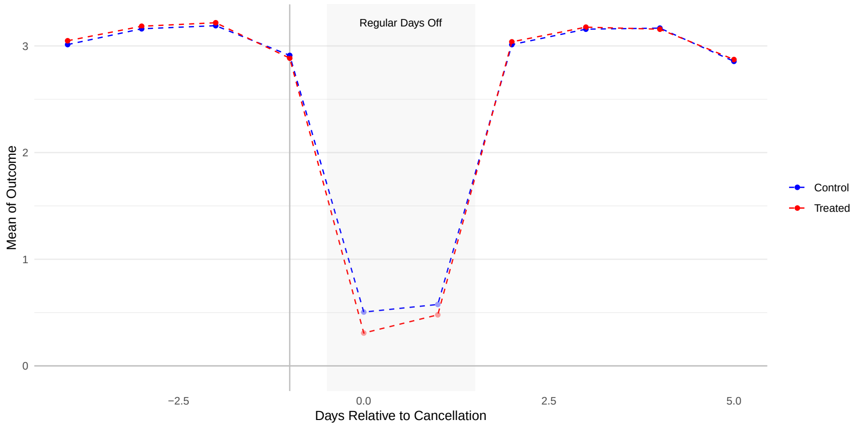


(c) Raw Data: Calls in Sick

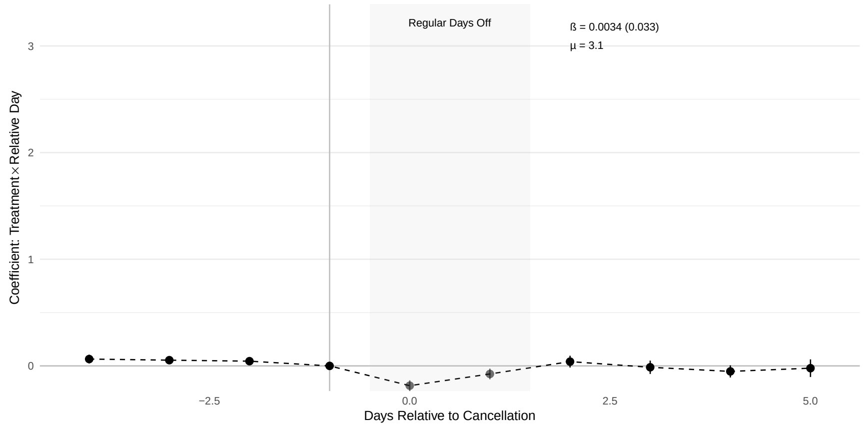


(d) Event Study: Calls in Sick

Figure 2: Effects on Officer Wellness



(a) Raw Data: Calls for Service



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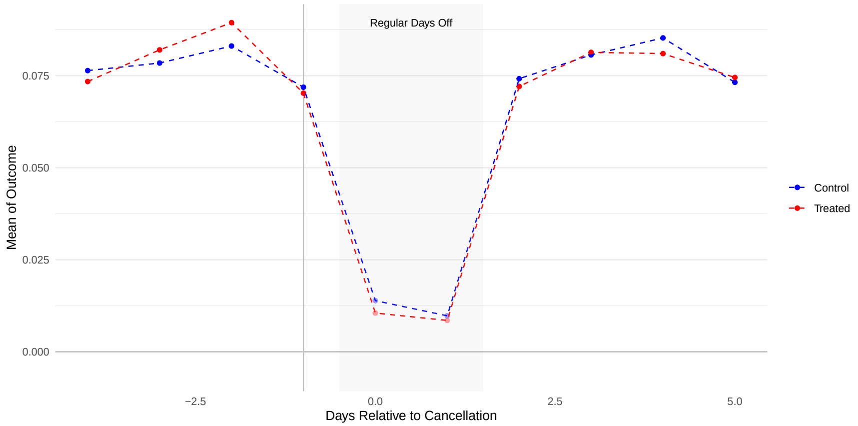
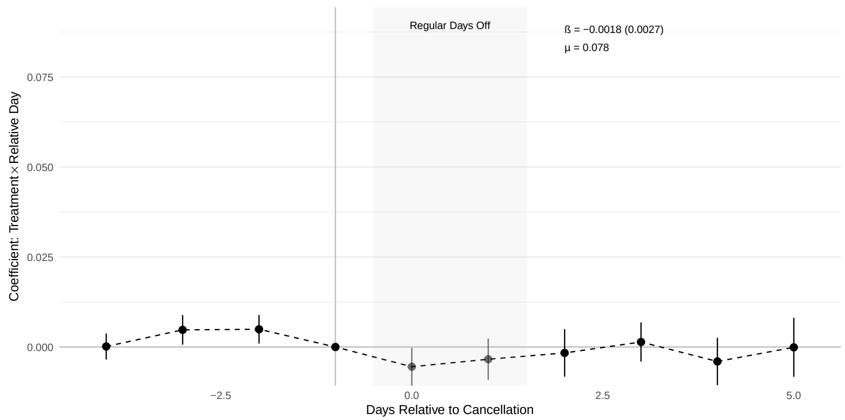
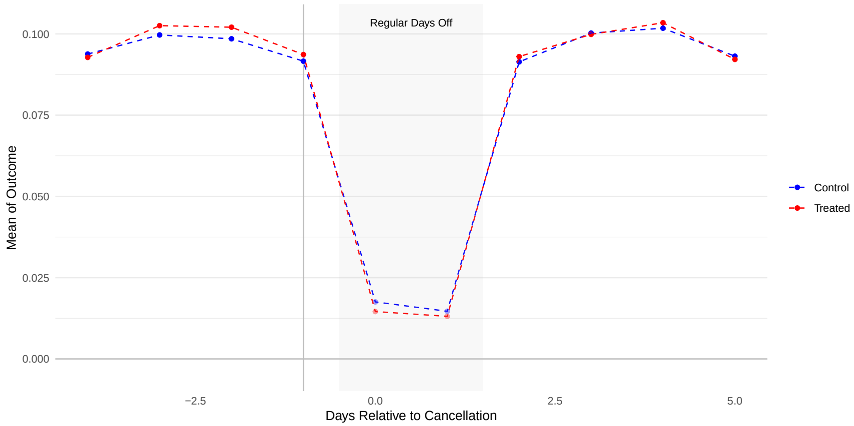


(c) Raw Data: Stops

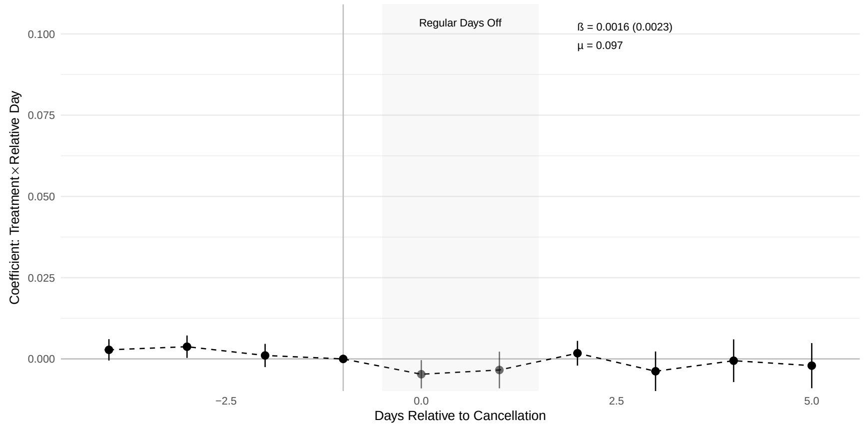


(d) Event Study: Stops

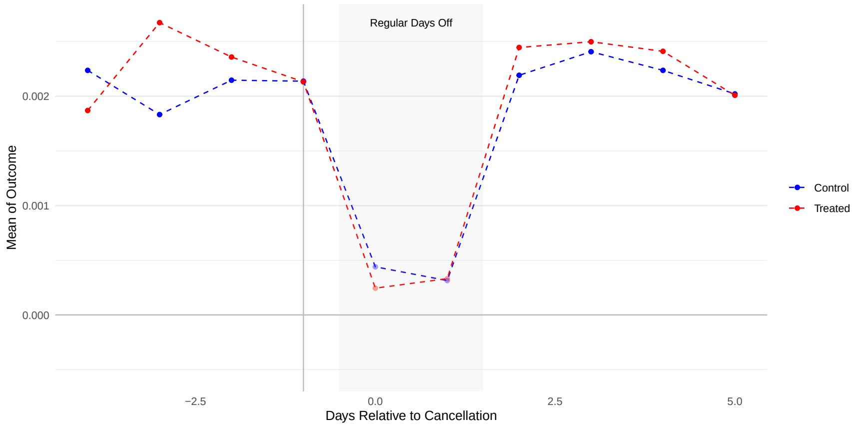
Figure 3: Effects on Productivity



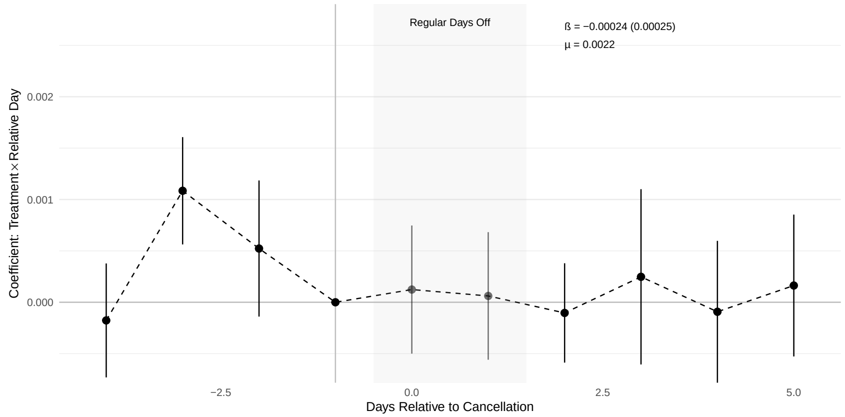
(a) Raw Data: Arrests



(b) Event Study: Arrests



(c) Raw Data: Use of Force



(d) Event Study: Use of Force

Figure 4: Effects on Law Enforcement Activity

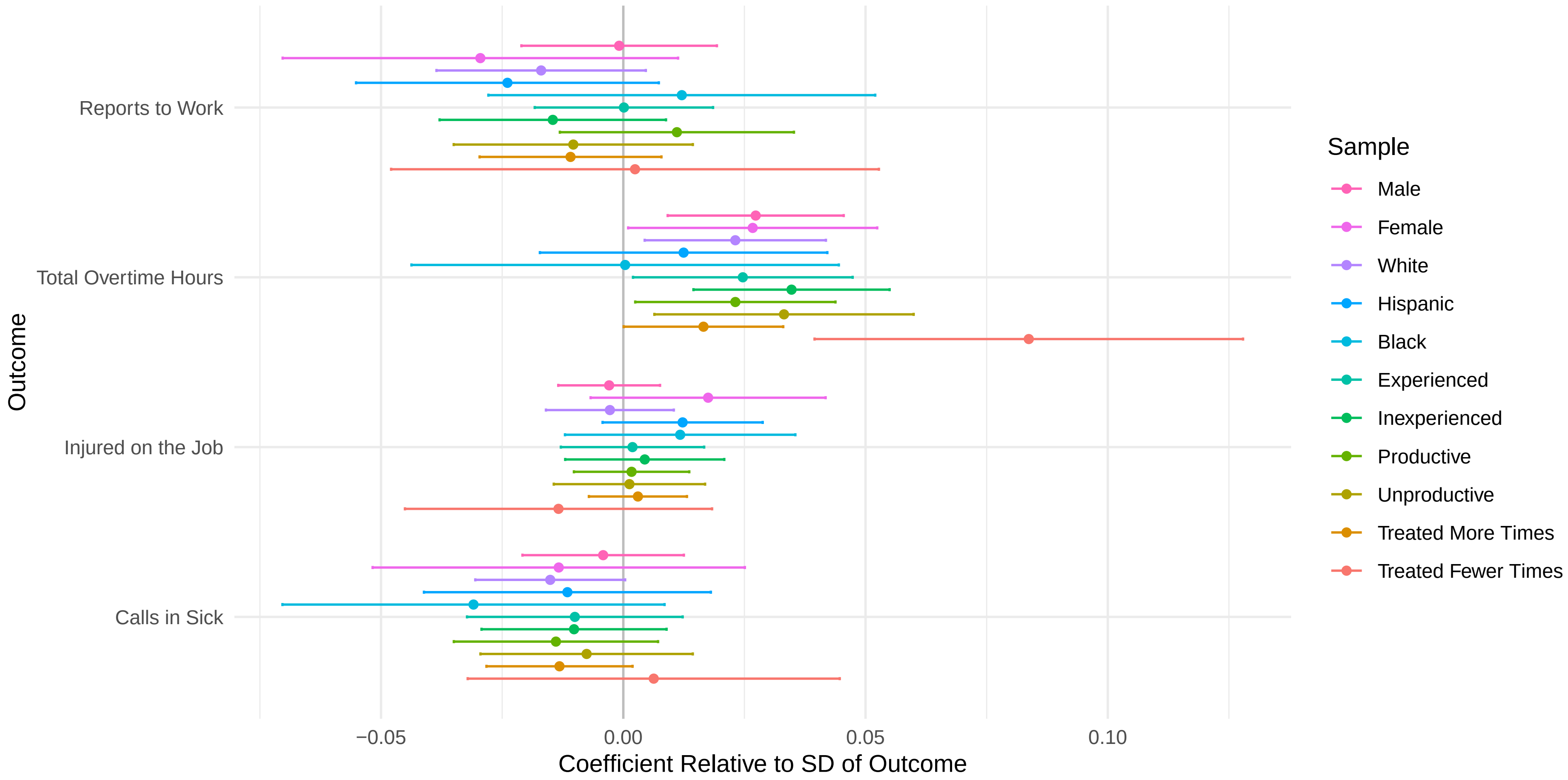


Figure 5: Attendance and Wellness

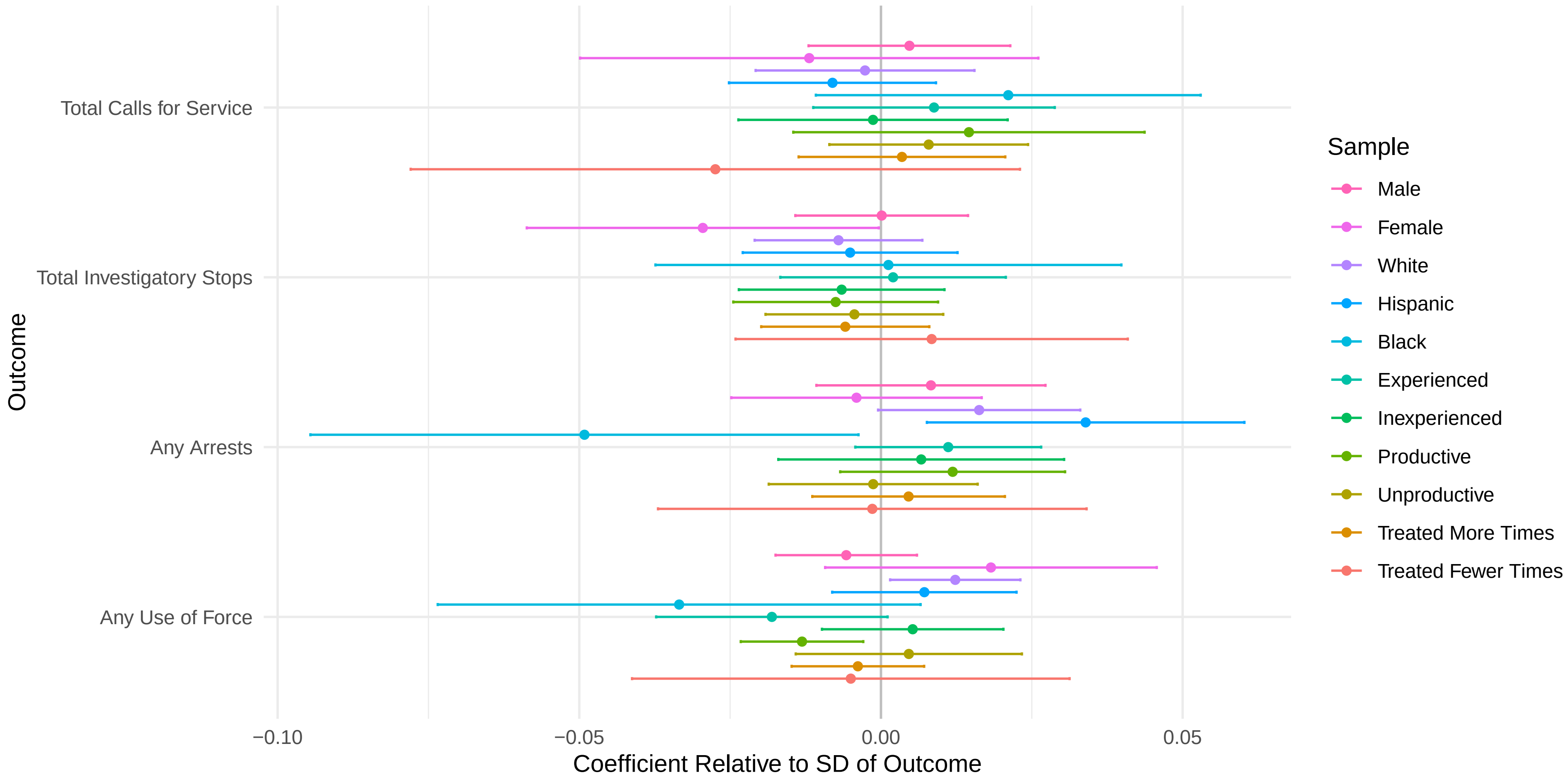


Figure 6: Enforcement and Productivity

A Data

We analyze data from CPD officers in DOGs 61-66, which includes approximately 66% of officer-date combinations we observe in a linked A&A and officer roster data set. We restrict our sample to this subset of DOGs because this

We also restrict our sample to officers assigned areas numbered 1-5. While The CPD does assigns some officers to "areas" with number labels greater than 5, these assignments are mainly administrative, are not associated with a specific geographic area, and so the job tasks fall outside our productivity measures of interest.

We identify RDO cancellations using cancellation matrices obtained from the CPD. Figure X shows an example of an RDO cancellation matrix we obtained from the CPD through a FOIA request. Figure X shows the sequential, cyclic structure to RDO cancellations, with occasional exceptions. For example, on May 31, 2022, the cancellation cycle begins with officers in Area 1 and DOG 61[10]. Comparing this figure to the operations calendar for 2022 in Figure X, it can be seen that May 31st is the first of two regularly scheduled days-off for DOG 61. The following day, June 1st, 2022, officers in Area 1 in DOG 62 were assigned an RDO cancellation and so on until by June 5th, 2022, all officers in DOGs 61 to 66 in Area 1 had been assigned treatment. Next, on June 7th, officers in all DOGs 61 to 66 in Areas 3 and 4 were sequentially assigned treatment, starting with DOG 62.

[10]DOG 75 also experiences an RDO cancellation on this date, but our analysis is restricted to officers in DOGs 61 to 66. See 2.2 for details on this sample restriction.

B Tables

Table B.4: Main Sample: No Officer Fixed-Effects

	(A) Attendance and Work		***(B) Officer Wellness***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
Treatment × Post	-0.0015	0.0412	-0.0006	-0.0026
	(0.0040)	(0.0142)	(0.0014)	(0.0024)
Pre-Period Mean	0.6735	0.5040	0.0444	0.0671
Observations	1348608	1348608	1345779	1345779
	(C) Officer Productivity		***(D) Enforcement Activity***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
Treatment × Post	0.0121	-0.0036	0.0016	-0.0001
	(0.0291)	(0.0032)	(0.0023)	(0.0002)
Pre-Period Mean	3.0738	0.0779	0.0965	0.0021
Observations	1348608	1348608	1348608	1348608

Table B.5: Main Sample: Including Area-DOG Fixed-Effects

	(A) Attendance and Work		***(B) Officer Wellness***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
Treatment × Post	-0.0034	0.0433	0.0002	-0.0018
	(0.0039)	(0.0141)	(0.0010)	(0.0018)
Pre-Period Mean	0.6735	0.5040	0.0444	0.0671
Observations	1348608	1348608	1345779	1345779
	(C) Officer Productivity		***(D) Enforcement Activity***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
Treatment × Post	0.0006	-0.0015	0.0014	-0.0001
	(0.0331)	(0.0026)	(0.0023)	(0.0002)
Pre-Period Mean	3.0738	0.0779	0.0965	0.0021
Observations	1348608	1348608	1348608	1348608

Table B.6: Main Results: Restricted Sample

	(A) Attendance and Work		***(B) Officer Wellness***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
Treatment × Post	-0.0070	0.0503	-0.0004	0.0022
	(0.0043)	(0.0150)	(0.0021)	(0.0027)
Pre-Period Mean	0.6727	0.5075	0.0443	0.0663
Observations	1028264	1028264	1026227	1026227
	(C) Officer Productivity		***(D) Enforcement Activity***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
Treatment × Post	0.0601	-0.0126	0.0040	-0.0002
	(0.0364)	(0.0048)	(0.0034)	(0.0003)
Pre-Period Mean	3.0658	0.0785	0.0966	0.0022
Observations	1028264	1028264	1028264	1028264

Table B.7: Arrest Type: Restricted Sample

	(1)	(2)	(3)	(4)
Dependent Variable:	Any Index Arrests	Any Traffic Arrests	Other Arrests	Any Arrests with No Charge
Treatment × Post	-0.0009	0.0006	0.0062	0.0002
	(0.0013)	(0.0016)	(0.0025)	(0.0006)
Pre-Period Mean	0.0195	0.0285	0.0836	0.0037
Observations	1028264	1028264	1028264	1028264

Table B.8: Main Results: Long Sample

	(A) Attendance and Work		***(B) Officer Wellness***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
Treatment × Post	-0.0107	0.0380	-0.0016	-0.0026
	(0.0063)	(0.0196)	(0.0017)	(0.0020)
Pre-Period Mean	0.6733	0.5575	0.0408	0.0642
Observations	1061472	1061472	1060274	1060274
	(C) Officer Productivity		***(D) Enforcement Activity***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
Treatment × Post	0.0149	0.0019	0.0048	-0.0005
	(0.0488)	(0.0049)	(0.0034)	(0.0003)
Pre-Period Mean	3.0096	0.0780	0.0959	0.0021
Observations	1061472	1061472	1061472	1061472

Table B.9: Arrest Type: Long Sample

	(1)	(2)	(3)	(4)
Dependent Variable:	Any Index Arrests	Any Traffic Arrests	Other Arrests	Any Arrests with No Charge
Treatment × Post	-0.0010	-0.0002	0.0050	-0.0009
	(0.0019)	(0.0023)	(0.0029)	(0.0008)
Pre-Period Mean	0.0197	0.0288	0.0826	0.0035
Observations	1061472	1061472	1061472	1061472

Table B.10: Pre-Trend Tests: Main Sample & Main Specification

	(A) Attendance and Work		*(B) Officer Wellness)*	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	302.0397321	201.0551106	1.5209373	1.9976566
p-value	0.0000000	0.0000000	0.2300015	0.1363351
Pre-Period Treatment Mean	0.6747454	0.4747561	0.0453578	0.0682885
Pre-Period Treatment SD	0.4684709	1.6002385	0.2080882	0.2522408
Pre-Period Control Mean	0.6727975	0.5190969	0.0438571	0.0665327
Pre-Period Control SD	0.4691924	1.6833528	0.2047774	0.2492113
Wtd. Correlation	0.5109903	0.5553233	0.1379648	0.5455009
	(C) Officer Productivity		*(D) Enforcement Activity)*	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	20.2627273	6.7768444	3.9595284	6.3529986
p-value	0.0000003	0.0013336	0.0175704	0.0019156
Pre-Period Treatment Mean	3.0839550	0.0787444	0.0977437	0.0022581
Pre-Period Treatment SD	4.1651260	0.4153973	0.2969684	0.0474656
Pre-Period Control Mean	3.0685397	0.0774226	0.0958958	0.0020883
Pre-Period Control SD	4.1620972	0.4180367	0.2944486	0.0456497
Wtd. Correlation	0.3104602	0.4527190	0.4266351	0.0120421

Table B.11: Pre-Trend Tests: Main Sample & Simple Specification

	(A) Attendance and Work		*(B) Officer Wellness)*	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	313.1556967	200.5580865	0.8598756	1.3633584
p-value	0.0000000	0.0000000	0.4729501	0.2735333
Pre-Period Treatment Mean	0.6747454	0.4747561	0.0453578	0.0682885
Pre-Period Treatment SD	0.4684709	1.6002385	0.2080882	0.2522408
Pre-Period Control Mean	0.6727975	0.5190969	0.0438571	0.0665327
Pre-Period Control SD	0.4691924	1.6833528	0.2047774	0.2492113
Wtd. Correlation	0.5109903	0.5553233	0.1379648	0.5455009
	(C) Officer Productivity		*(D) Enforcement Activity)*	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	19.4663838	6.9752308	4.0845952	6.4866391
p-value	0.0000004	0.0011289	0.0155293	0.0017073
Pre-Period Treatment Mean	3.0839550	0.0787444	0.0977437	0.0022581
Pre-Period Treatment SD	4.1651260	0.4153973	0.2969684	0.0474656
Pre-Period Control Mean	3.0685397	0.0774226	0.0958958	0.0020883
Pre-Period Control SD	4.1620972	0.4180367	0.2944486	0.0456497
Wtd. Correlation	0.3104602	0.4527190	0.4266351	0.0120421

Table B.12: Pre-Trend Tests: Main Sample & Area-DOG Fixed-Effects Specification

	(A) Attendance and Work		***(B) Officer Wellness)***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	300.7312180	200.8699125	0.9356466	2.1296576
p-value	0.0000000	0.0000000	0.4360869	0.1180822
Pre-Period Treatment Mean	0.6747454	0.4747561	0.0453578	0.0682885
Pre-Period Treatment SD	0.4684709	1.6002385	0.2080882	0.2522408
Pre-Period Control Mean	0.6727975	0.5190969	0.0438571	0.0665327
Pre-Period Control SD	0.4691924	1.6833528	0.2047774	0.2492113
Wtd. Correlation	0.5109903	0.5553233	0.1379648	0.5455009
	(C) Officer Productivity		***(D) Enforcement Activity)***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	20.8148879	6.4463934	4.0561482	6.3696465
p-value	0.0000002	0.0017673	0.0159704	0.0018882
Pre-Period Treatment Mean	3.0839550	0.0787444	0.0977437	0.0022581
Pre-Period Treatment SD	4.1651260	0.4153973	0.2969684	0.0474656
Pre-Period Control Mean	3.0685397	0.0774226	0.0958958	0.0020883
Pre-Period Control SD	4.1620972	0.4180367	0.2944486	0.0456497
Wtd. Correlation	0.3104602	0.4527190	0.4266351	0.0120421

Table B.13: Pre-Trend Tests: Restricted Sample & Main Specification

	(A) Attendance and Work		***(B) Officer Wellness)***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	247.1326974	210.0559918	3.2054616	3.0593382
p-value	0.0000000	0.0000000	0.0376942	0.0438641
Pre-Period Treatment Mean	0.6747383	0.4747682	0.0453674	0.0682678
Pre-Period Treatment SD	0.4684736	1.6002645	0.2081090	0.2522054
Pre-Period Control Mean	0.6711392	0.5337314	0.0434106	0.0647226
Pre-Period Control SD	0.4698001	1.7068395	0.2037800	0.2460361
Wtd. Correlation	0.4670226	0.5012863	0.0758335	0.5214857
	(C) Officer Productivity		***(D) Enforcement Activity)***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	8.9876642	1.4283707	4.4030739	6.3363712
p-value	0.0002292	0.2546609	0.0113842	0.0019433
Pre-Period Treatment Mean	3.0835707	0.0787565	0.0977511	0.0022542
Pre-Period Treatment SD	4.1650315	0.4154349	0.2969784	0.0474248
Pre-Period Control Mean	3.0515069	0.0782578	0.0956617	0.0021492
Pre-Period Control SD	4.1449496	0.4234372	0.2941272	0.0463096
Wtd. Correlation	0.2592356	0.3532044	0.3643425	-0.0072551

Table B.14: Pre-Trend Tests: Restricted Sample & Simple Specification

	(A) Attendance and Work		***(B) Officer Wellness)***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	248.9478864	208.3513903	0.5243370	2.5781648
p-value	0.0000000	0.0000000	0.6690064	0.0728470
Pre-Period Treatment Mean	0.6747383	0.4747682	0.0453674	0.0682678
Pre-Period Treatment SD	0.4684736	1.6002645	0.2081090	0.2522054
Pre-Period Control Mean	0.6711392	0.5337314	0.0434106	0.0647226
Pre-Period Control SD	0.4698001	1.7068395	0.2037800	0.2460361
Wtd. Correlation	0.4670226	0.5012863	0.0758335	0.5214857
	(C) Officer Productivity		***(D) Enforcement Activity)***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	6.7335530	3.6520707	5.2013041	6.7078462
p-value	0.0013833	0.0238924	0.0053579	0.0014138
Pre-Period Treatment Mean	3.0835707	0.0787565	0.0977511	0.0022542
Pre-Period Treatment SD	4.1650315	0.4154349	0.2969784	0.0474248
Pre-Period Control Mean	3.0515069	0.0782578	0.0956617	0.0021492
Pre-Period Control SD	4.1449496	0.4234372	0.2941272	0.0463096
Wtd. Correlation	0.2592356	0.3532044	0.3643425	-0.0072551

Table B.15: Pre-Trend Tests: Restricted Sample & Area-DOG Fixed-Effects Specification

	(A) Attendance and Work		***(B) Officer Wellness)***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	247.3805156	210.6727322	2.4165828	2.8022744
p-value	0.0000000	0.0000000	0.0866029	0.0574315
Pre-Period Treatment Mean	0.6747383	0.4747682	0.0453674	0.0682678
Pre-Period Treatment SD	0.4684736	1.6002645	0.2081090	0.2522054
Pre-Period Control Mean	0.6711392	0.5337314	0.0434106	0.0647226
Pre-Period Control SD	0.4698001	1.7068395	0.2037800	0.2460361
Wtd. Correlation	0.4670226	0.5012863	0.0758335	0.5214857
	(C) Officer Productivity		***(D) Enforcement Activity)***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	9.2273004	1.2020315	4.4888492	6.3318641
p-value	0.0001916	0.3265005	0.0104811	0.0019509
Pre-Period Treatment Mean	3.0835707	0.0787565	0.0977511	0.0022542
Pre-Period Treatment SD	4.1650315	0.4154349	0.2969784	0.0474248
Pre-Period Control Mean	3.0515069	0.0782578	0.0956617	0.0021492
Pre-Period Control SD	4.1449496	0.4234372	0.2941272	0.0463096
Wtd. Correlation	0.2592356	0.3532044	0.3643425	-0.0072551

Table B.16: Pre-Trend Tests: Long Sample & Main Specification

	(A) Attendance and Work		*(B) Officer Wellness)*	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	95.7650200	118.2689214	0.9233558	2.0756918
p-value	0.0000000	0.0000000	0.5320330	0.0568625
Pre-Period Treatment Mean	0.6754153	0.5493703	0.0394068	0.0639287
Pre-Period Treatment SD	0.4682204	1.7281597	0.1945611	0.2446263
Pre-Period Control Mean	0.6713546	0.5653076	0.0421151	0.0643839
Pre-Period Control SD	0.4697216	1.7551157	0.2008521	0.2454361
Wtd. Correlation	0.4435911	0.5390457	-0.0860767	0.4520714
	(C) Officer Productivity		*(D) Enforcement Activity)*	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	5.0210799	2.4064782	3.3343893	3.4796196
p-value	0.0002347	0.0287684	0.0045679	0.0034687
Pre-Period Treatment Mean	2.9796416	0.0789532	0.0981926	0.0022150
Pre-Period Treatment SD	4.0488436	0.4228886	0.2975754	0.0470122
Pre-Period Control Mean	3.0384712	0.0770828	0.0937103	0.0020540
Pre-Period Control SD	4.1336424	0.4183387	0.2914258	0.0452742
Wtd. Correlation	0.2268424	0.2095894	0.3161691	-0.0052038

Table B.17: Pre-Trend Tests: Long Sample & Simple Specification

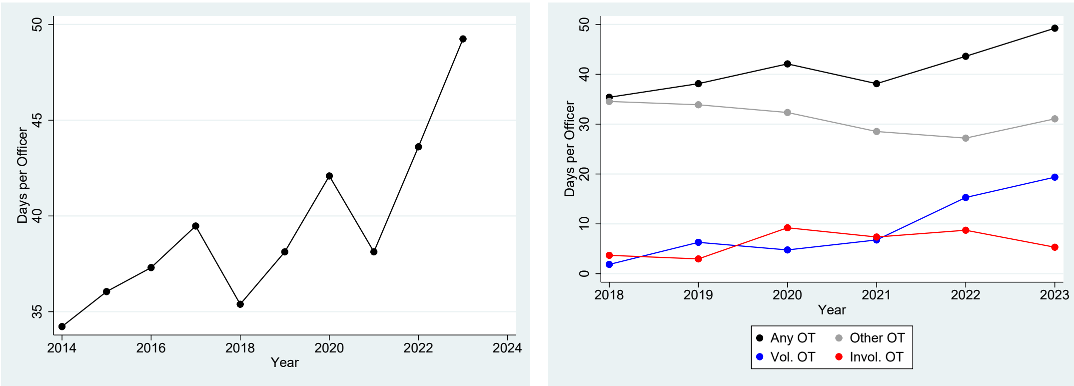
	(A) Attendance and Work		*(B) Officer Wellness)*	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	96.3337044	116.0085614	1.2132132	1.6825751
p-value	0.0000000	0.0000000	0.3221592	0.1278518
Pre-Period Treatment Mean	0.6754153	0.5493703	0.0394068	0.0639287
Pre-Period Treatment SD	0.4682204	1.7281597	0.1945611	0.2446263
Pre-Period Control Mean	0.6713546	0.5653076	0.0421151	0.0643839
Pre-Period Control SD	0.4697216	1.7551157	0.2008521	0.2454361
Wtd. Correlation	0.4435911	0.5390457	-0.0860767	0.4520714
	(C) Officer Productivity		*(D) Enforcement Activity)*	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	4.1990988	1.5498580	3.0503162	3.7245403
p-value	0.0009375	0.1673381	0.0079098	0.0021991
Pre-Period Treatment Mean	2.9796416	0.0789532	0.0981926	0.0022150
Pre-Period Treatment SD	4.0488436	0.4228886	0.2975754	0.0470122
Pre-Period Control Mean	3.0384712	0.0770828	0.0937103	0.0020540
Pre-Period Control SD	4.1336424	0.4183387	0.2914258	0.0452742
Wtd. Correlation	0.2268424	0.2095894	0.3161691	-0.0052038

Table B.18: Pre-Trend Tests: Long Sample & Area-DOG Fixed-Effects Specification

	(A) Attendance and Work		***(B) Officer Wellness)***	
	(1)	(2)	(3)	(4)
Dependent Variable:	Reports to Work	Total Overtime Hours	Injured on the Job	Calls in Sick
F-stat	95.4408383	118.0609585	1.1059467	2.2950539
p-value	0.0000000	0.0000000	0.3914637	0.0361601
Pre-Period Treatment Mean	0.6754153	0.5493703	0.0394068	0.0639287
Pre-Period Treatment SD	0.4682204	1.7281597	0.1945611	0.2446263
Pre-Period Control Mean	0.6713546	0.5653076	0.0421151	0.0643839
Pre-Period Control SD	0.4697216	1.7551157	0.2008521	0.2454361
Wtd. Correlation	0.4435911	0.5390457	-0.0860767	0.4520714
	(C) Officer Productivity		***(D) Enforcement Activity)***	
	(5)	(6)	(7)	(8)
Dependent Variable:	Total Calls for Service	Total Investigatory Stops	Any Arrests	Any Use of Force
F-stat	5.1940922	2.4099247	3.3095029	3.4411353
p-value	0.0001778	0.0285662	0.0047903	0.0037299
Pre-Period Treatment Mean	2.9796416	0.0789532	0.0981926	0.0022150
Pre-Period Treatment SD	4.0488436	0.4228886	0.2975754	0.0470122
Pre-Period Control Mean	3.0384712	0.0770828	0.0937103	0.0020540
Pre-Period Control SD	4.1336424	0.4183387	0.2914258	0.0452742
Wtd. Correlation	0.2268424	0.2095894	0.3161691	-0.0052038

C Appendix Figures

C.1 Overtime Descriptive Statistics

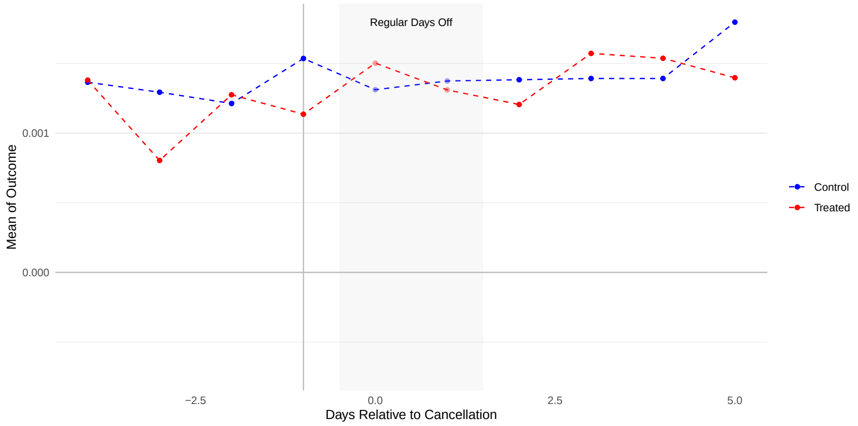


(a) All Overtime Since 2014

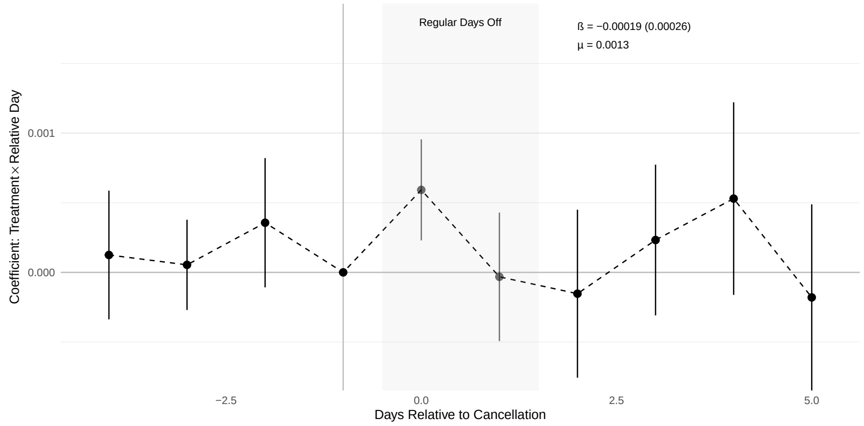
(b) By Overtime Type Since 2018

Figure C.7: Average Days Working Overtime per Officer

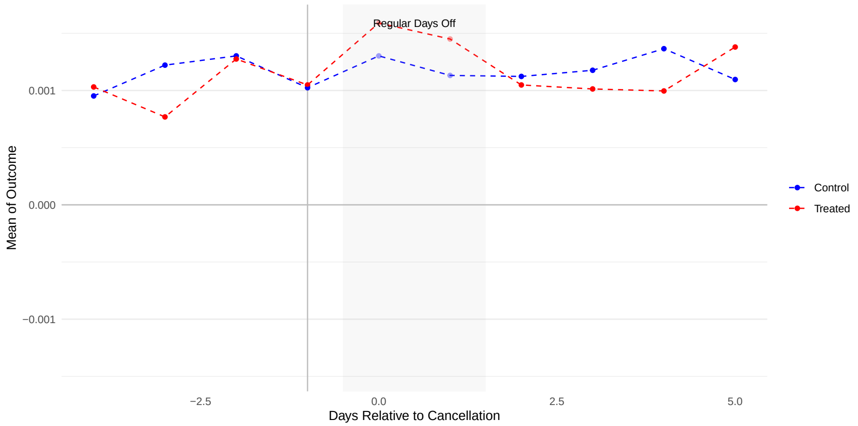
C.2 Compliance with Work Assignments



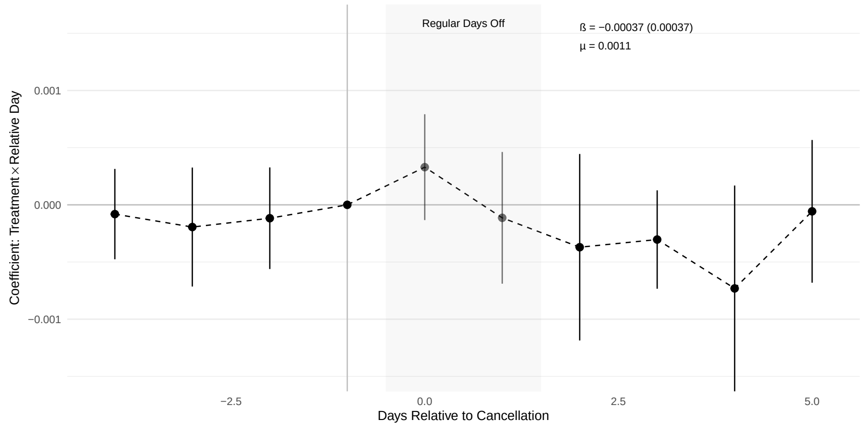
(a) Raw Data: DOG Assignment



(b) Event Study: DOG Assignment



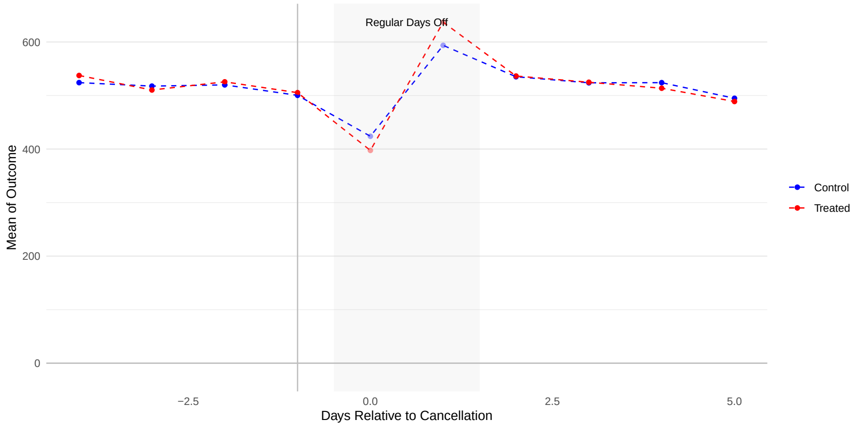
(c) Raw Data: Area Assignment



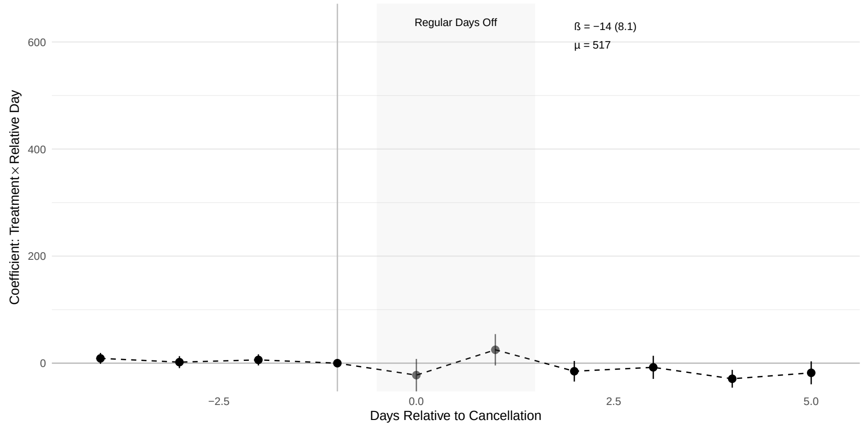
(d) Event Study: Area Assignment

Figure C.8: Change in Work Assignment

C.3 Call Response Time



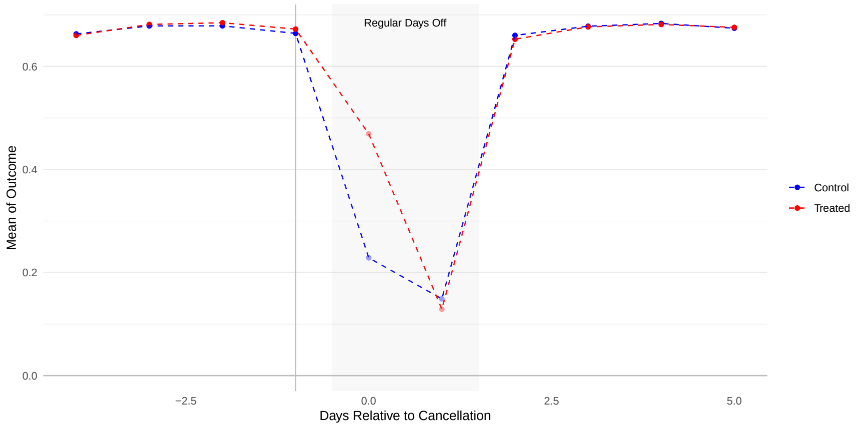
(a) Binscatter Plot



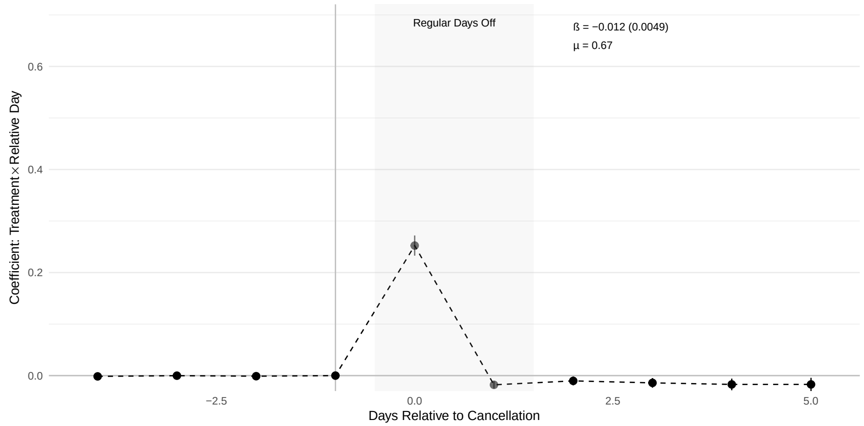
(b) Dynamic DID Event Study

Figure C.9: Time from Call Received by Dispatcher to On-Scene in Seconds

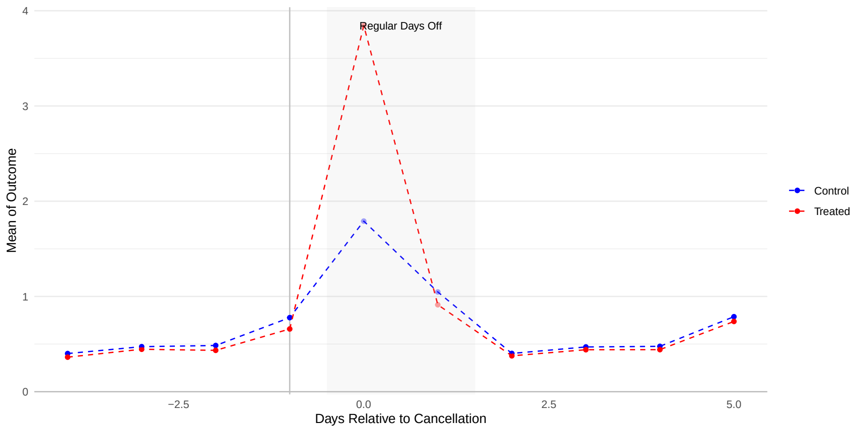
C.4 Event Study Plots: Restricted Sample



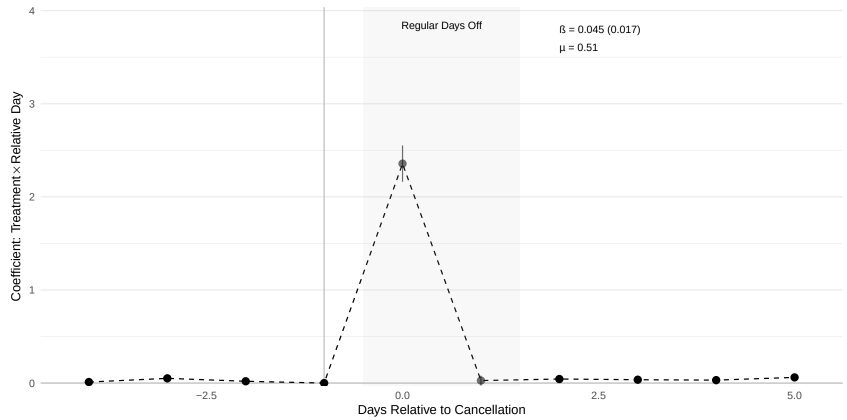
(a) Raw Data: At Work



(b) Event Study: At Work

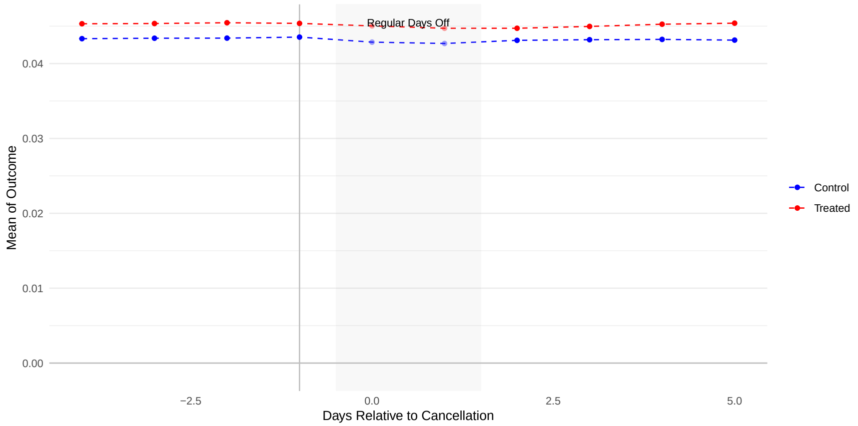


(c) Raw Data: Overtime Hours

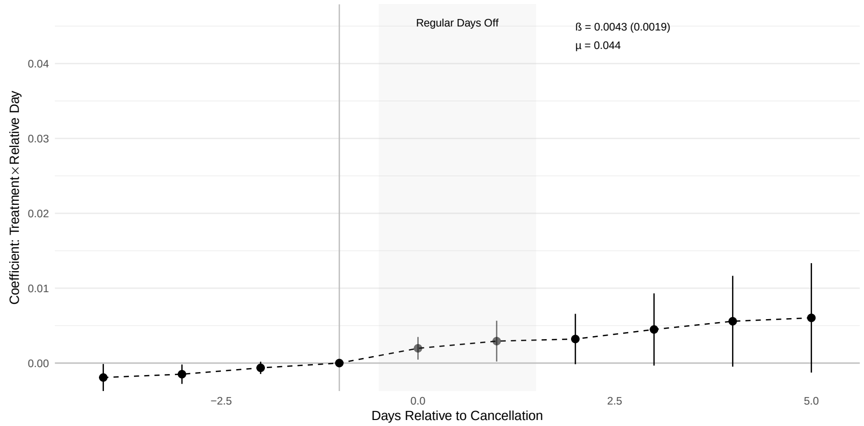


(d) Event Study: Overtime Hours

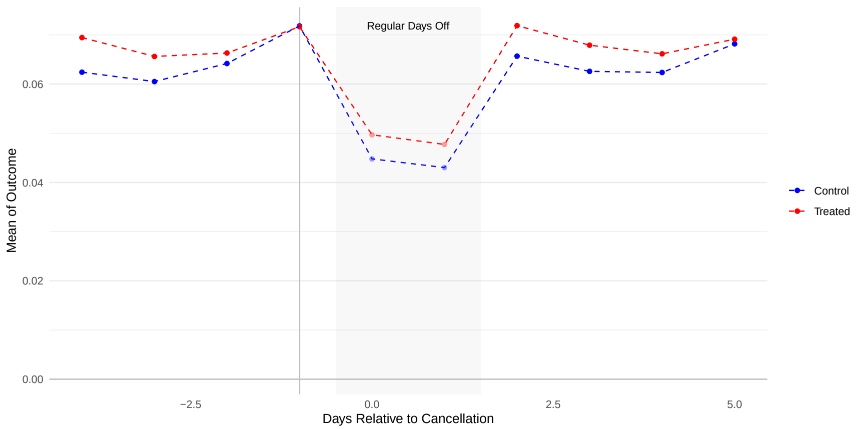
Figure C.10: Treatment Compliance and Effects on Attendance



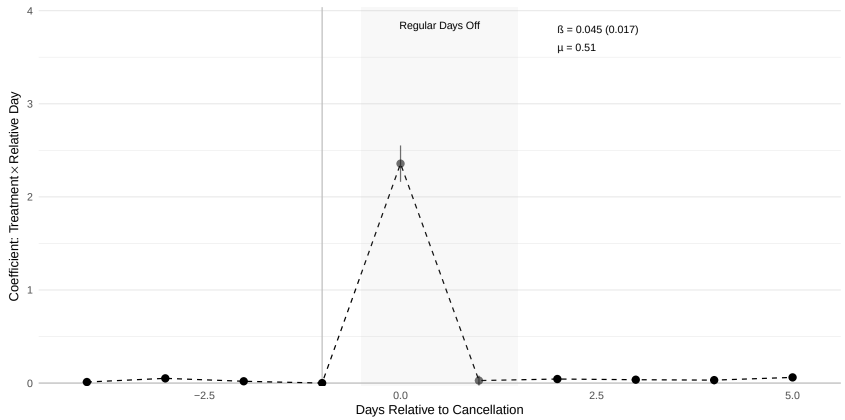
(a) Raw Data: Injury



(b) Event Study: Injury

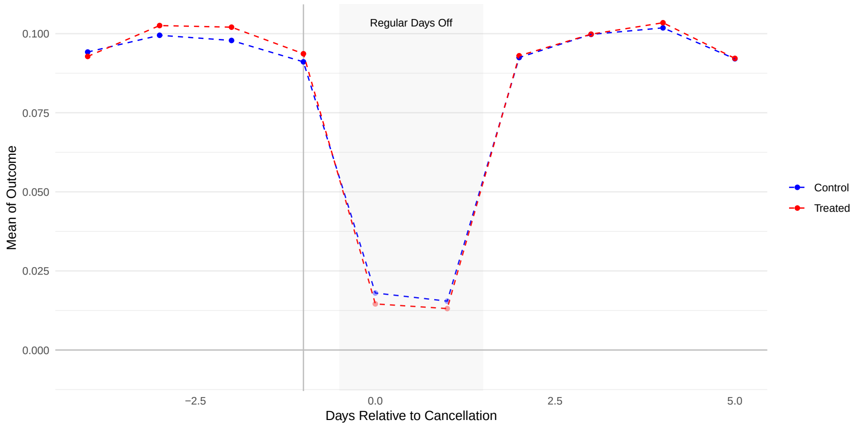


(c) Raw Data: Calls in Sick

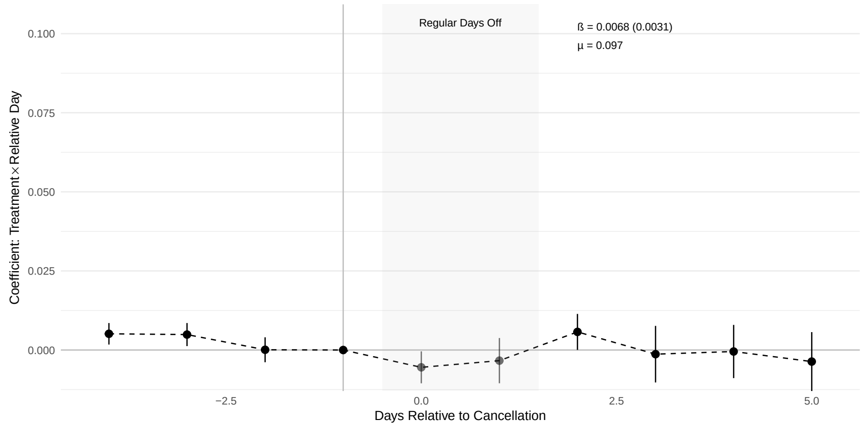


(d) Event Study: Calls in Sick

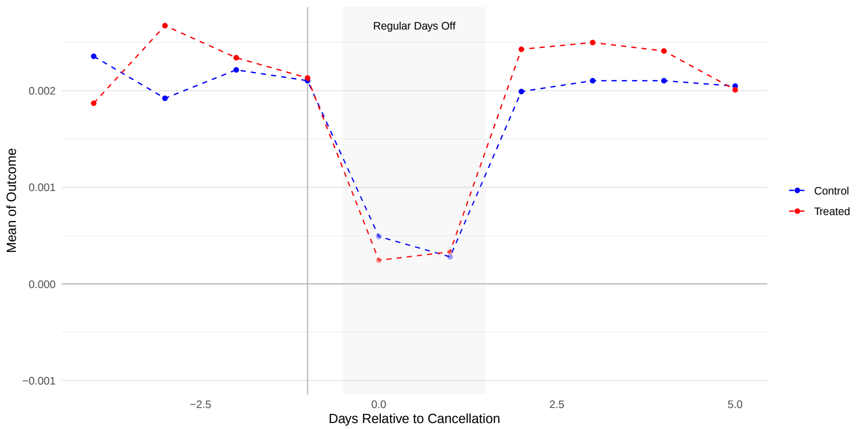
Figure C.11: Effects on Officer Wellness



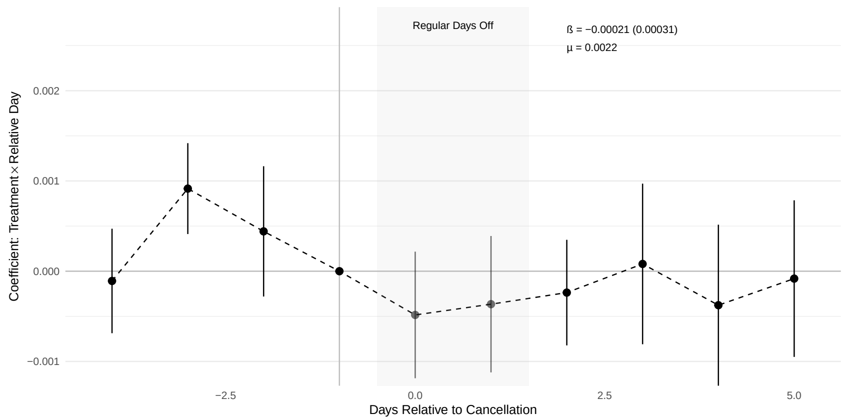
(a) Raw Data: Arrests



(b) Event Study: Arrests

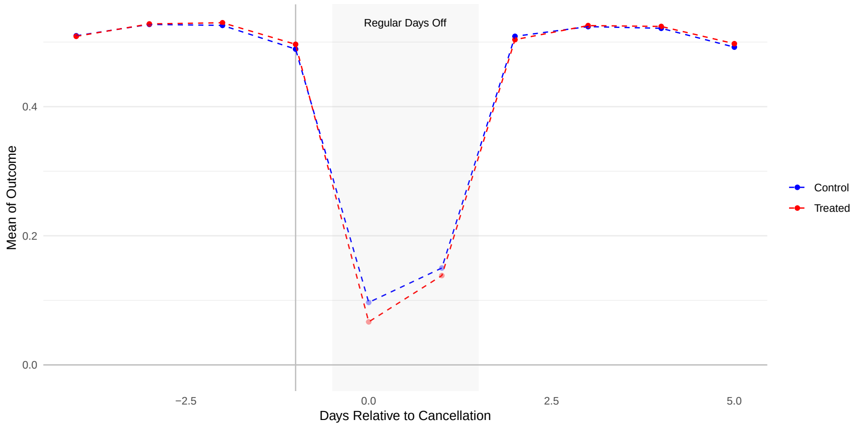


(c) Raw Data: Use of Force

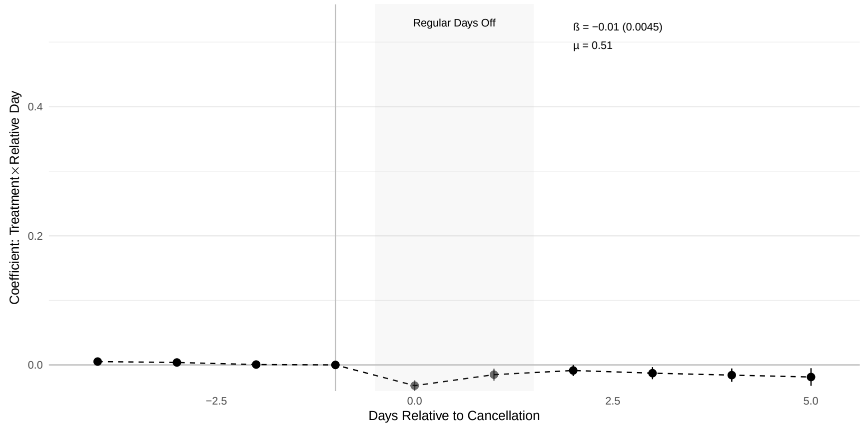


(d) Event Study: Use of Force

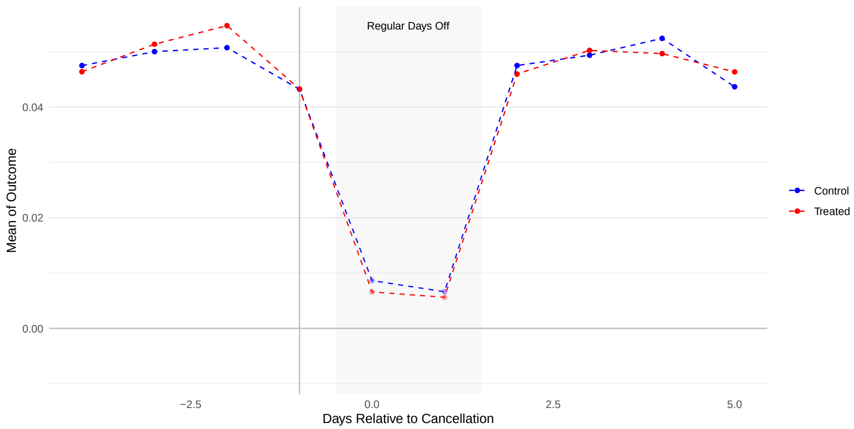
Figure C.12: Effects on Law Enforcement Activity



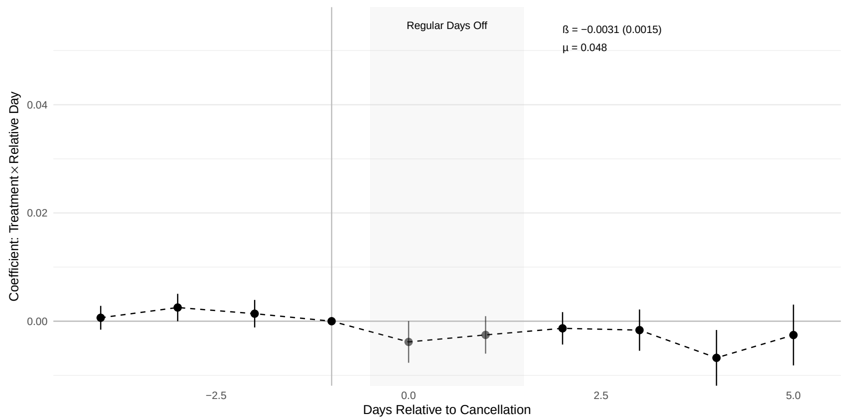
(a) Raw Data: Calls for Service



(b) Event Study: Calls for Service



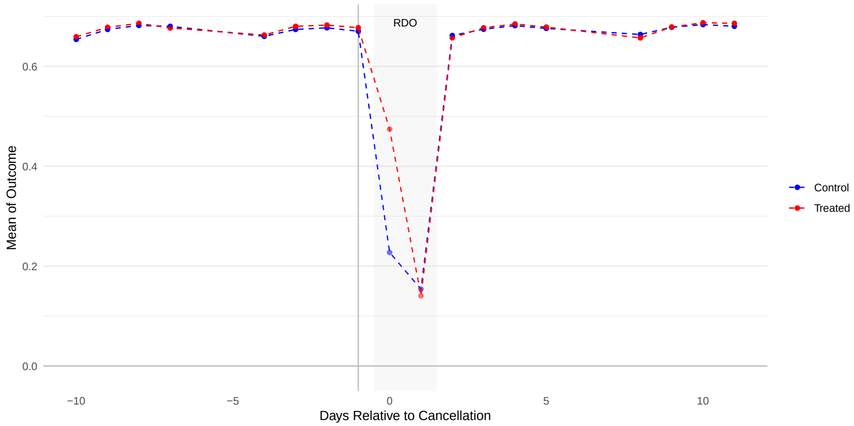
(c) Raw Data: Stops



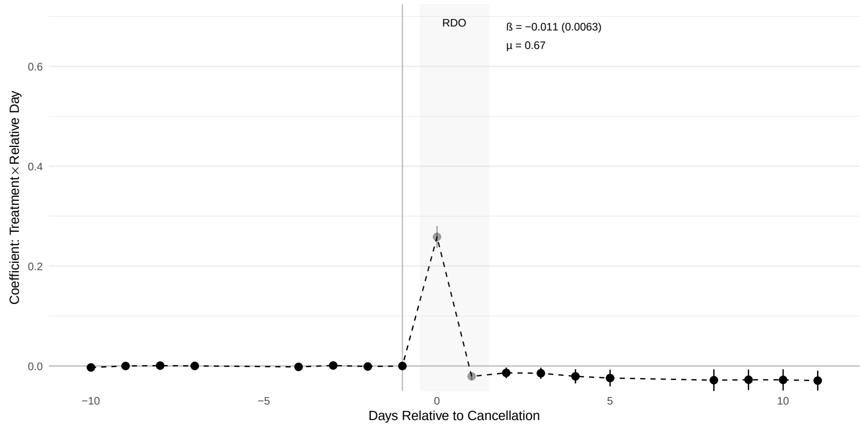
(d) Event Study: Stops

Figure C.13: Effects on Productivity

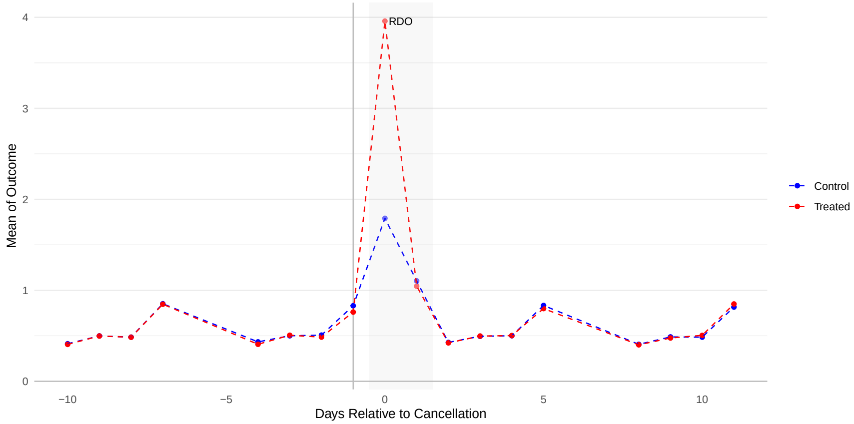
C.5 Event Study Plots: Long Sample



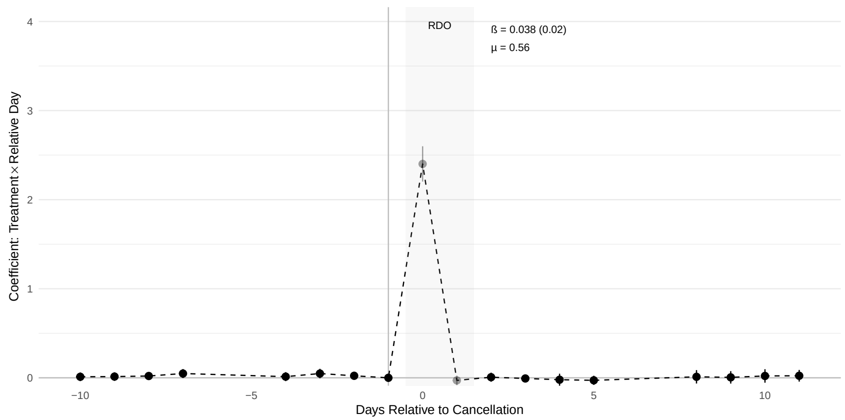
(a) Raw Data: At Work



(b) Event Study: At Work

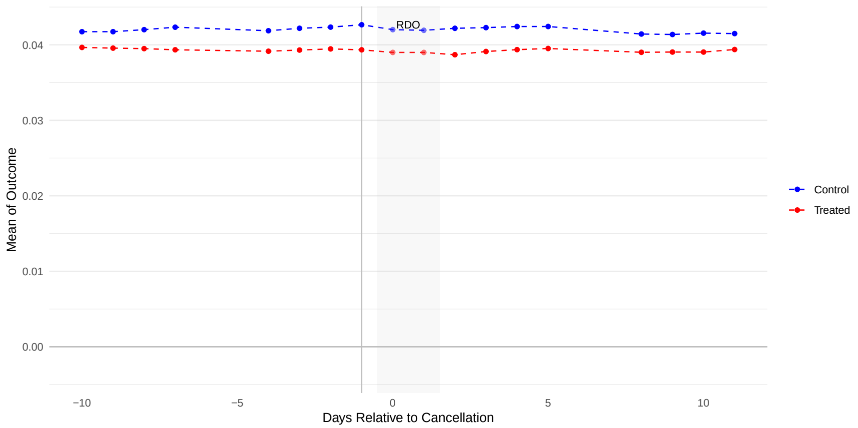


(c) Raw Data: Overtime Hours

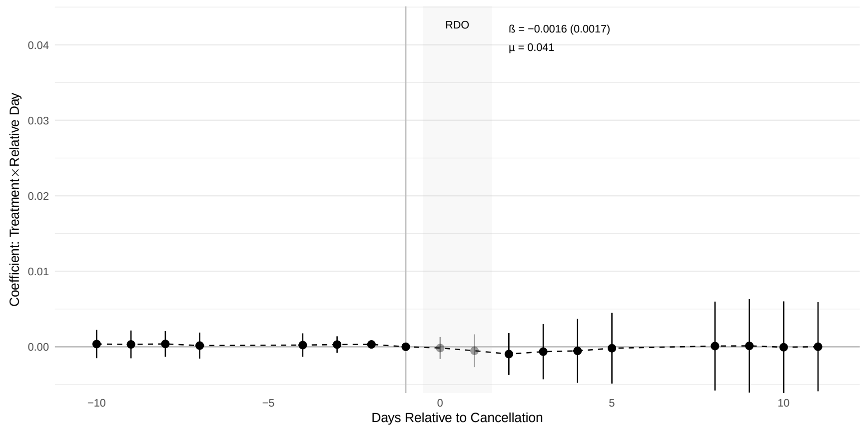


(d) Event Study: Overtime Hours

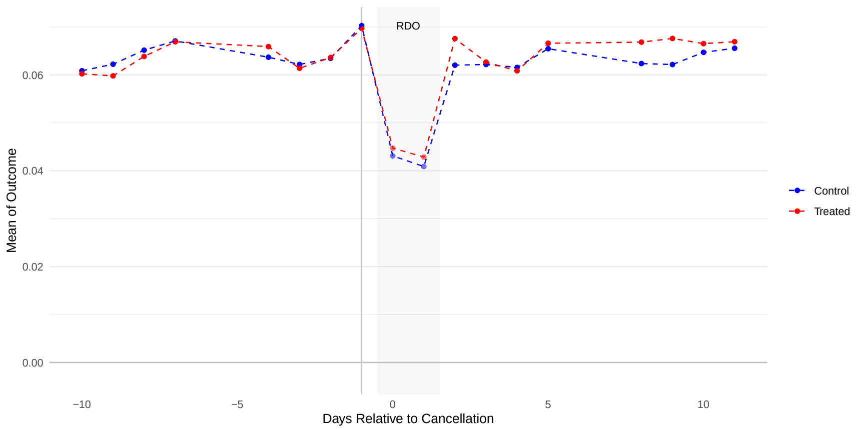
Figure C.14: Treatment Compliance and Effects on Attendance



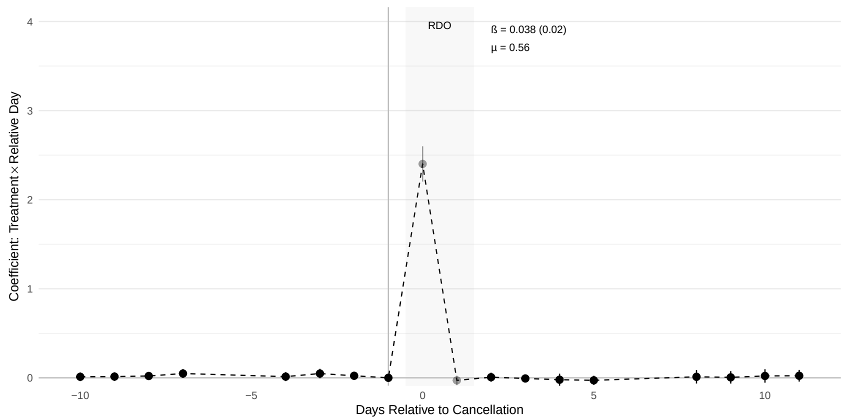
(a) Raw Data: Injury



(b) Event Study: Injury

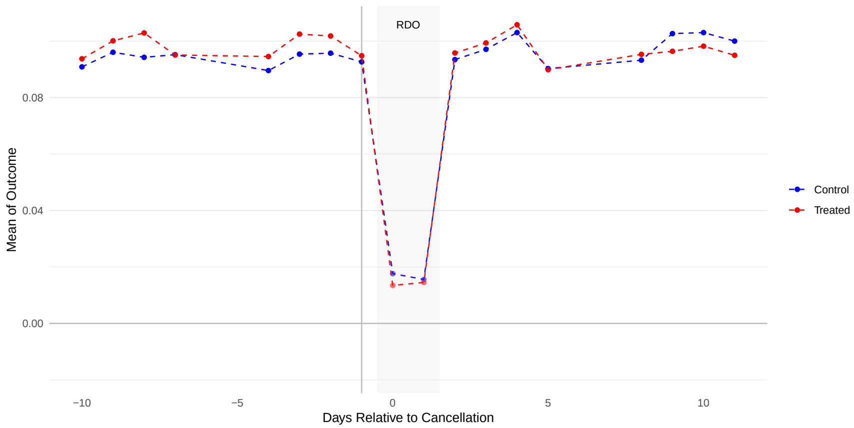


(c) Raw Data: Calls in Sick

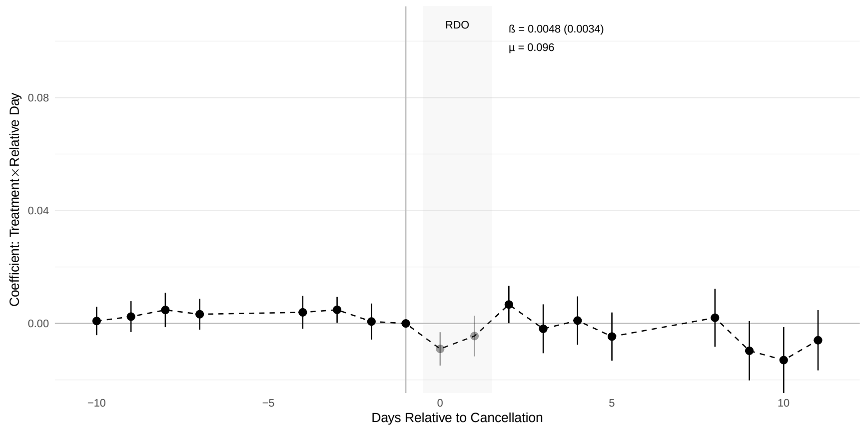


(d) Event Study: Calls in Sick

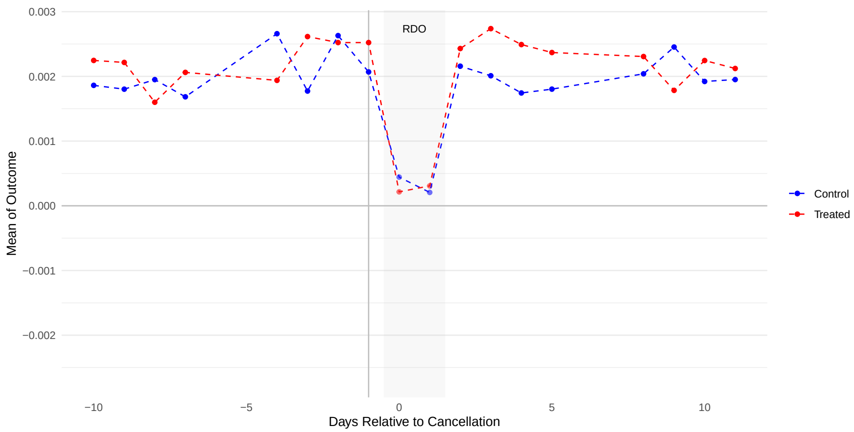
Figure C.15: Effects on Officer Wellness



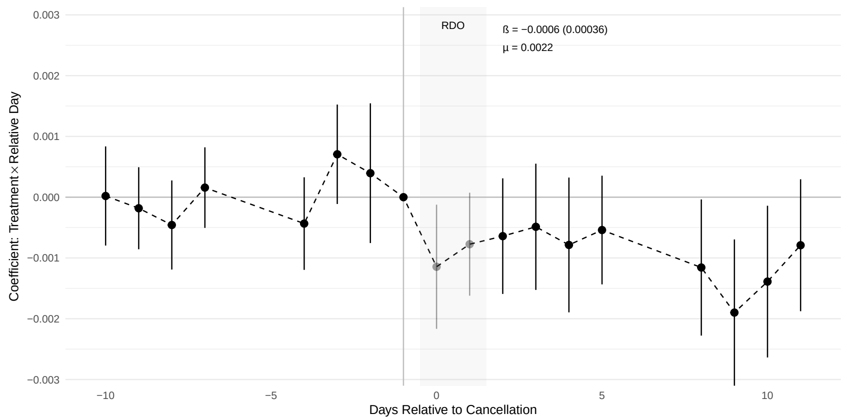
(a) Raw Data: Arrests



(b) Event Study: Arrests

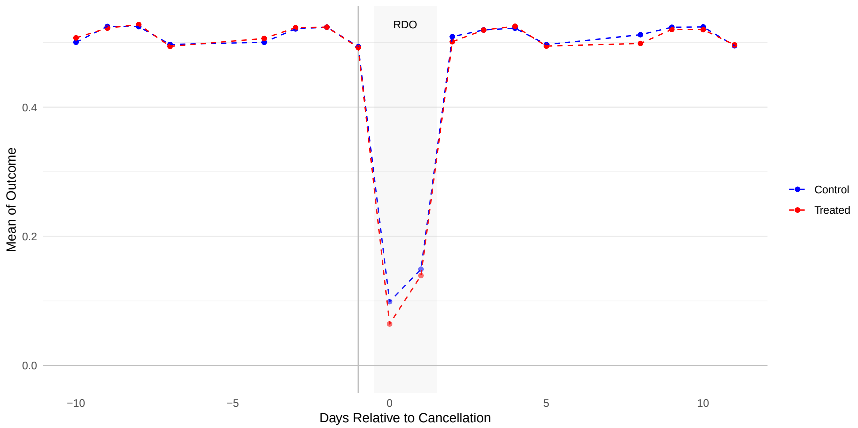


(c) Raw Data: Use of Force

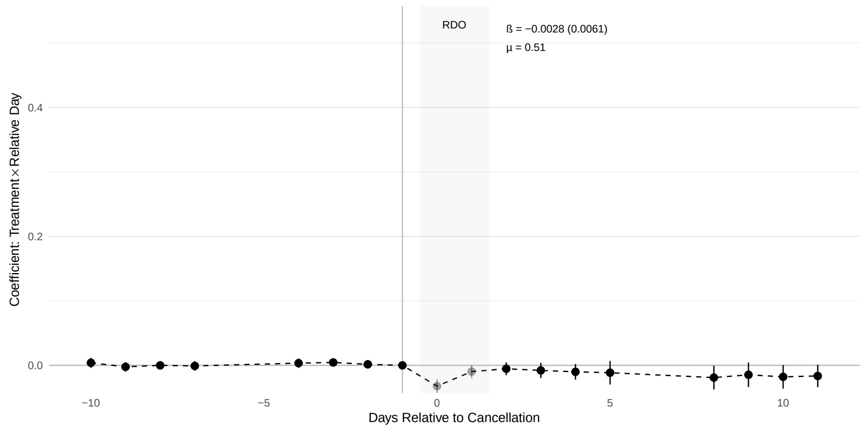


(d) Event Study: Use of Force

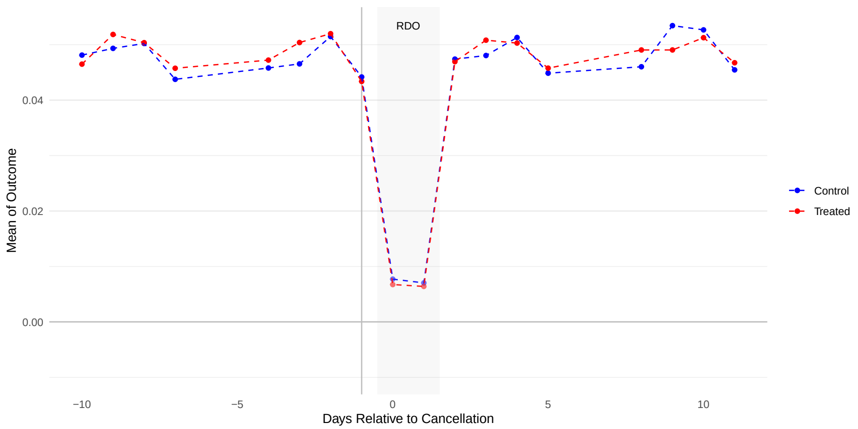
Figure C.16: Effects on Law Enforcement Activity



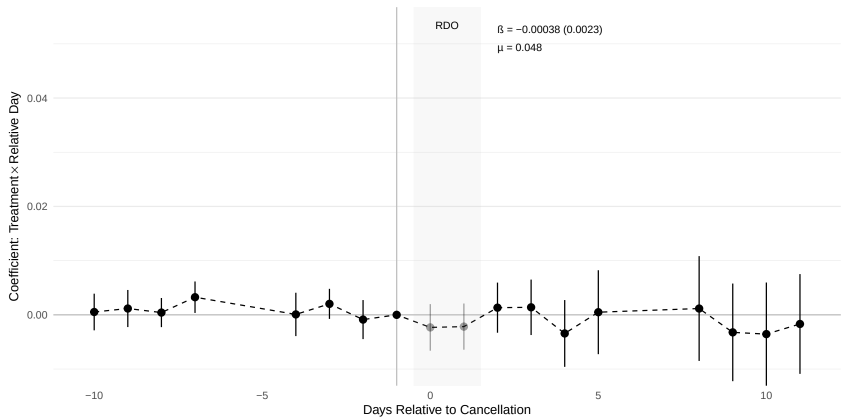
(a) Raw Data: Calls for Service



(b) Event Study: Calls for Service



(c) Raw Data: Stops



(d) Event Study: Stops

Figure C.17: Effects on Productivity